

Prehospital Real-Time AI for Trauma Mortality Prediction: A Multi-Institutional and Multi-National Validation Study

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This study applies machine learning algorithms to develop a predictive model for emergency department (ED) mortality due to trauma, using only 21 pre-hospital variables. The model was trained on a large-scale database in South Korea, and external validation was conducted using data from four domestic institutions and one institution in Australia.

According to Table 1, the incidence of ED mortality—the target outcome of this analysis—is extremely low: approximately 1% at domestic facilities in South Korea and 0.1% at the Australian institution. Since this analysis deals with highly imbalanced data, SMOTE was used during model training. However, the model's performance was evaluated using only AUROC, sensitivity, specificity, accuracy, and balanced accuracy, which is insufficient. When handling severely imbalanced data, AUROC tends to overestimate the model's utility, and thus AUPRC should be used for a more rigorous evaluation. Additionally, precision (positive predictive value) should be included as a threshold-dependent metric. Calibration plots should also be presented to evaluate the predictive performance, but none were included in this paper, raising doubts about the conclusion that the model is highly effective.

Regarding external validation, Table 3 shows high AUROC values at domestic facilities in South Korea but lower AUROC values for the Australian dataset. Although the data collection periods differ, this raises the possibility of overfitting, and the international generalizability of the model remains uncertain.

From a triage perspective, given the reported sensitivity and specificity, and considering the low outcome prevalence, the positive predictive value would be only a few percent, resulting in a high number of false positives. The actual benefit of using this predictive model is unclear. A Decision Curve Analysis would be useful in this context, but such an evaluation was not conducted.

(Remarks on code availability)

The code for model training and testing is adequate. However, since the data has not been shared and the trained model is not provided, it is difficult to verify the reproducibility of the predictions.

Reviewer #2

(Remarks to the Author)

The manuscript by Oh et al. presents a real-time AI model designed to predict emergency room (ER) mortality among pre-hospital trauma patients. The model was trained on retrospective data from the Korean Trauma Data Bank (KTDB) and externally validated using retrospective datasets from four regional trauma centers in South Korea and one Level 1 trauma center in Australia. The model demonstrated strong performance in both internal and external validations, outperforming the Shock Index. While the study offers a promising approach for early mortality prediction in trauma care, several major concerns need to be addressed:

1. Data Representation in Table 1: The data presentation in Table 1 is unclear, particularly for variables with categories

labeled as "uncheckable" or "unchecked." For instance, the SpO₂ (%) feature includes 3.0% as uncheckable and 51.2% as unchecked, alongside subcategories of 0.2%, 1.1%, 2.7%, 6.1%, and 86.9% for different value ranges. These proportions do not add up to 100%, leading to confusion about how the data is categorized and interpreted.

2. Model Interpretability: While SHAP values were used to identify the importance of input features in predicting mortality, the manuscript lacks a discussion on model interpretability. The authors should elaborate on how these key features relate to trauma mortality and provide clinical insights or implications for practice.

3. Comparative Evaluation: The model's performance is only compared against the Shock Index. However, as noted in the Introduction, other existing models (referenced in citations 6, 7, 8, and 13) also address similar tasks. A thorough comparison and discussion of how the proposed model performs relative to these existing approaches is necessary to better contextualize its contribution.

4. Reproducibility and Code Availability: Although the authors have shared scripts for data preprocessing, model training, and validation via a GitHub repository, important resources are missing. Specifically, detailed scripts and documentation of the data preprocessing steps, access to raw or sample datasets, and the trained model itself (and the web-based AI system for mobile use) are not provided. These omissions hinder the reproducibility of the study's results.

(Remarks on code availability)

Although the authors have shared scripts for data preprocessing, model training, and validation via a GitHub repository, important resources are missing. Specifically, detailed scripts and documentation of the data preprocessing steps, access to raw or sample datasets, and the trained model itself (and the web-based AI system for mobile use) are not provided. These omissions hinder the reproducibility of the study's results.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this revision, the evaluation using AUROC and precision, calibration, and decision curve analysis that were previously pointed out have been added, improving the interpretability of the results in the context of imbalanced data. The external validation using data from Korea appears to be generally robust, and the DCA results also suggest clinical usefulness.

On the other hand, for the Australian dataset, although the AUROC appears favorable at 0.895, the AUPRC is quite low at 0.064. The DCA shows a marginally higher net benefit, but only within a very narrow range of thresholds, and the overall net benefit is considerably lower compared to that based on Korean data. This is likely due to the influence of missing values and differences in emergency medical systems between countries. Given that some features used in the prediction model inherently reflect country-specific system characteristics, this limitation is unavoidable and also indicates that the generalizability of the model to other countries is limited.

Therefore, as mentioned at the beginning of the Discussion, it would be inaccurate to state that the model performed well even for the Australian data. A more appropriate interpretation would be that while the external validation using Korean data yielded robust results, the model's usefulness for the Australian dataset remains limited due to differences in healthcare systems and data collection processes.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

The authors have addressed most of the concerns I raised in the previous round. However, I am unable to access the GitHub repository and the web server, which are intended to provide the code and model for application.

(Remarks on code availability)

I am unable to access the GitHub repository and the web server, which are intended to provide the code and model for application.

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thank you for your thorough and considerate responses to the review comments. I have no additional revisions to request at

this time.

(Remarks on code availability)

Nothing especially.

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Response Letter

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We would like to express our sincere gratitude to the reviewers for their thoughtful and constructive feedback on our manuscript. We carefully reviewed each comment and have revised the manuscript accordingly.

In response to Reviewer #1's valuable suggestions, we incorporated additional evaluation metrics suitable for imbalanced data, including AUPRC and precision, and we now provide calibration plots and decision curve analysis to better demonstrate the model's clinical utility. We have also addressed concerns regarding external validation and generalizability by clarifying differences in data sources and further discussing potential limitations.

In response to Reviewer #2's comments, we have clarified the data representation in Table 1 and revised the relevant descriptions to improve interpretability. We expanded our discussion on model interpretability using SHAP values, providing clinical insights into key predictors. We also added further discussion of other relevant models previously cited in the manuscript to better contextualize the contribution of our proposed approach. To enhance reproducibility, we have updated our GitHub repository with detailed documentation of preprocessing steps, the trained model, the web-based inference system, and a sample dataset.

We deeply appreciate the reviewers' efforts in improving the clarity, rigor, and impact of our work. Below, we provide a point-by-point response to each comment.

We have carefully revised the document to make the reviewer's comments and our responses clearly distinguishable. As requested, our responses are marked in blue text, and changes in the manuscript are highlighted in red for clarity. In addition, page numbers have been added to facilitate easy cross-referencing.

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

This study applies machine learning algorithms to develop a predictive model for emergency department (ED) mortality due to trauma, using only 21 pre-hospital variables. The model was trained on a large-scale database in South Korea, and external validation was conducted using data from four domestic institutions and one institution in Australia.

According to Table 1, the incidence of ED mortality—the target outcome of this analysis—is extremely low: approximately 1% at domestic facilities in South Korea and 0.1% at the Australian institution. Since this analysis deals with highly imbalanced data, SMOTE was used during model training. However, the model's performance was evaluated using only AUROC, sensitivity, specificity, accuracy, and balanced accuracy, which is insufficient. When handling severely imbalanced data, AUROC tends to overestimate the model's utility, and thus AUPRC should be used for a more rigorous evaluation. Additionally, precision (positive predictive value) should be included as a threshold-dependent metric. Calibration plots should also be presented to evaluate the predictive performance, but none were included in this paper, raising doubts about the conclusion that the model is highly effective.

Response> We appreciate the reviewer's insightful comment. As noted, our dataset exhibited a highly imbalanced outcome distribution, which presented a considerable challenge during the analysis. In accordance with the reviewer's recommendation, we conducted additional evaluations including precision, AUPRC, and calibration plots. Furthermore, given the imbalanced nature of the dataset, we incorporated additional evaluation metrics tailored for rare-event prediction. In addition to the Lift metric—which quantifies how much better the model performs compared to random selection by normalizing precision against the baseline prevalence of positive cases—we also included the adjusted precision and the adjusted precision-recall gain metrics. The latter adjusts the precision-recall curve by

accounting for class imbalance and expressing model gains relative to a random classifier, thereby enhancing interpretability in imbalanced scenarios. Moreover, we reported the normalized area under the precision-recall curve (normalized AUPRC), which facilitates fairer comparisons across models and datasets with varying class imbalance. Collectively, these metrics provide a more comprehensive assessment of model performance in highly imbalanced settings. Through this extended analysis, we found that our original model demonstrated suboptimal calibration performance. To address this, we applied a post hoc calibration procedure using the beta calibration method based on the beta distribution [DOI: 10.1214/17-EJS1338SI]. Beta calibration is a post-processing method used to improve the probability estimates from binary classifiers. It extends logistic calibration by using a more flexible transformation of predicted probabilities. This allows the method to better correct distorted or asymmetric probability outputs. Model parameters are fitted using maximum likelihood on a separate validation dataset. The recalibrated model demonstrated improved performance, achieving a calibration slope of 0.982 and a Brier score of 0.0069, as follows:

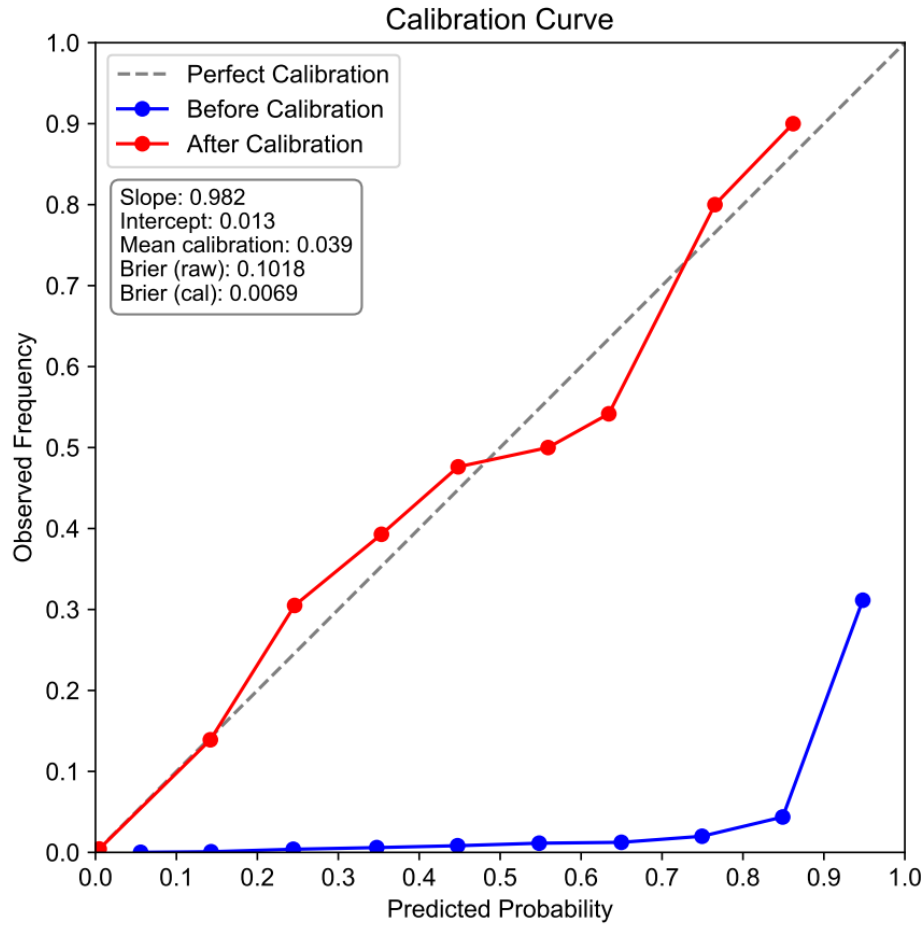


Figure 3. Calibration plot of the proposed model before and after calibration

Figure 3 illustrates the calibration performance of our prediction model before and after applying the beta calibration method. The blue line represents the original model's output, which shows poor alignment with the ideal (diagonal) line, indicating suboptimal calibration. In contrast, the red line depicts the model's output after beta calibration, demonstrating a close fit to the diagonal and indicating improved alignment between predicted probabilities and observed outcomes. Quantitatively, the recalibrated model achieved a calibration slope of 0.982 and an intercept of 0.013, both suggesting good calibration. The Brier score improved significantly from 0.1018 (uncalibrated) to 0.0069 (calibrated), and the mean calibration error was reduced to 0.039. These results confirm that the post hoc beta calibration method effectively enhanced the model's reliability in probability estimation.

We added in the method section as follows:

“Model Calibration

We applied a post hoc calibration procedure using the beta calibration method based on the beta distribution [DOI: 10.1214/17-EJS1338SI]. Beta calibration is a post-processing method used to improve the probability estimates from binary classifiers. It extends logistic calibration by using a more flexible transformation of predicted probabilities. This allows the method to better correct distorted or asymmetric probability outputs. Model parameters are fitted using maximum likelihood on a separate validation dataset.”

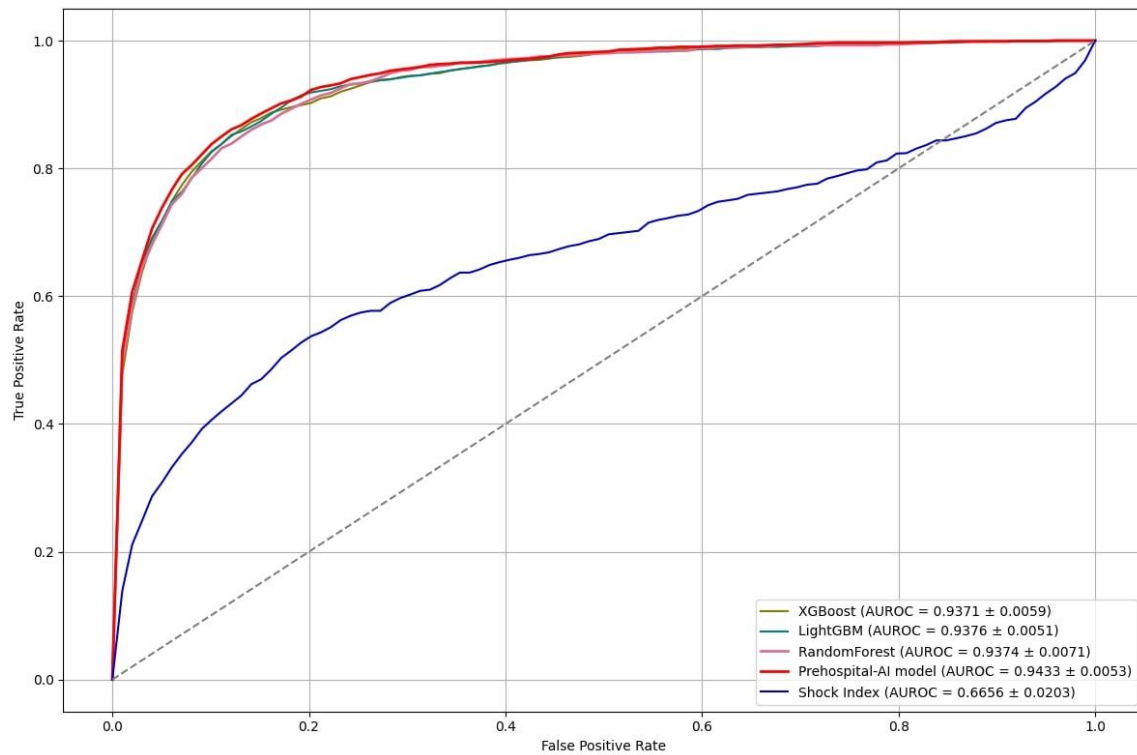
We also added in the result section as follows:

“Model Calibration

We found that our original model demonstrated suboptimal calibration performance. The recalibrated model demonstrated improved performance, achieving a calibration slope of 0.982 and a Brier score of 0.0069. **Figure 3** illustrates the calibration performance of our prediction model before and after applying the beta calibration method. The blue line represents the original model's output, which shows poor alignment with the ideal (diagonal) line, indicating suboptimal calibration. In contrast, the red line depicts the model's output after beta calibration, demonstrating a close fit to the diagonal and indicating improved alignment between predicted probabilities and observed outcomes. Quantitatively, the recalibrated model achieved a calibration slope of 0.982 and an intercept of 0.013, both suggesting good calibration. The Brier score improved significantly from 0.1018 (uncalibrated) to 0.0069 (calibrated), and the mean calibration error was reduced to 0.039. These results confirm that the post hoc beta calibration method effectively enhanced the model's reliability in probability estimation. Figure 4 illustrates the performance of the AI models after calibration.”

We added the AUPRC figure and revised the original figure 2 which showed performance of 5-fold cross validation as follows:

(A)



(B)

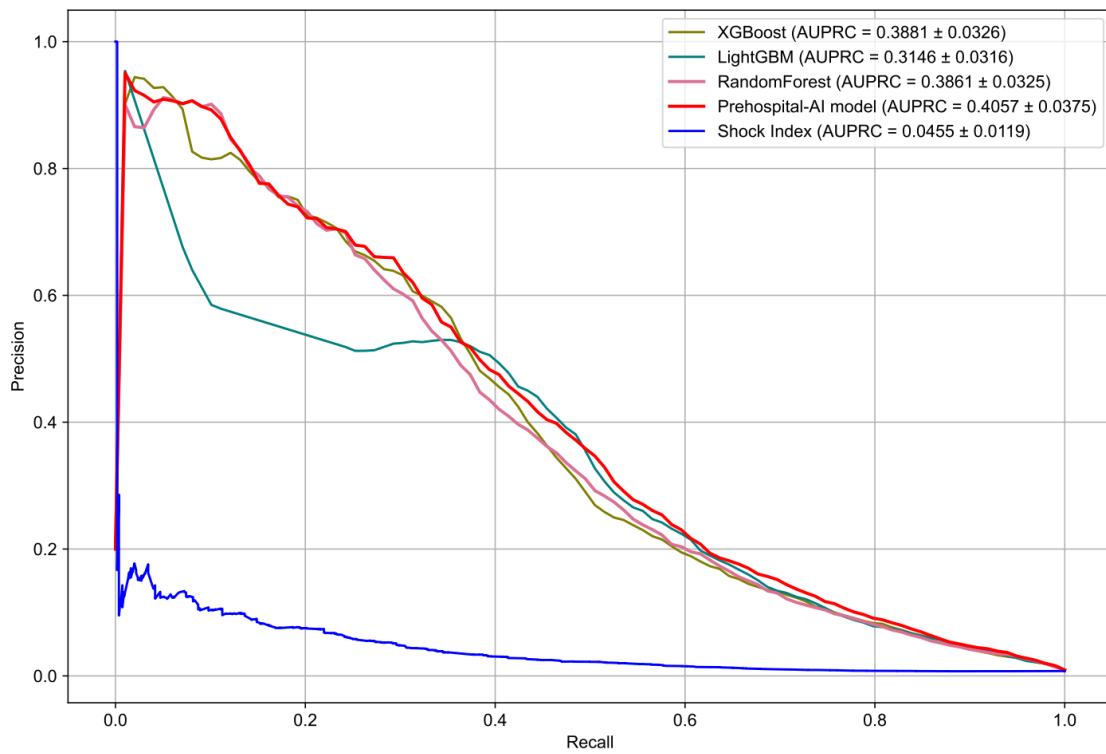
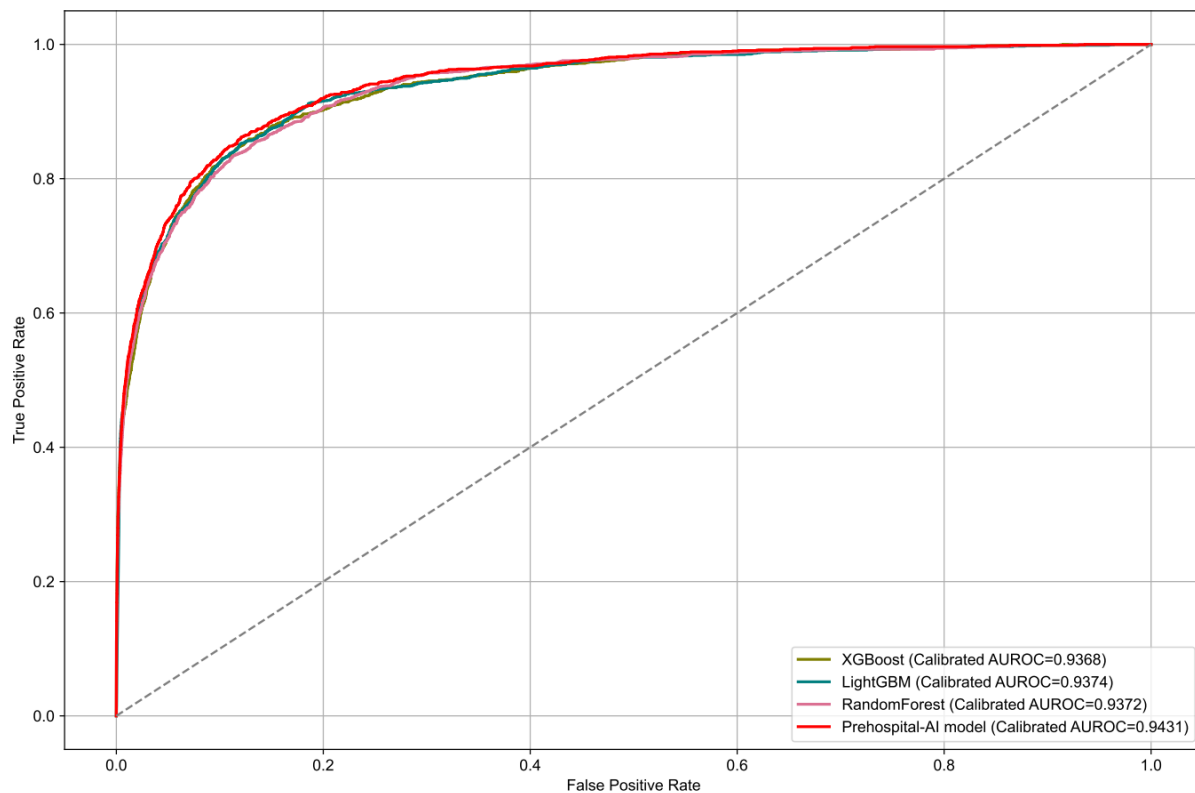


Figure 2. AUROC (A) and AUPRC (B) performance of the top three individual models, the Prehospital-AI ensemble, and the Shock Index based on 5-fold cross-validation.

We added new Figure 4 which illustrates AUROC and AUPRC performance after calibration as follows:

(A)



(B)

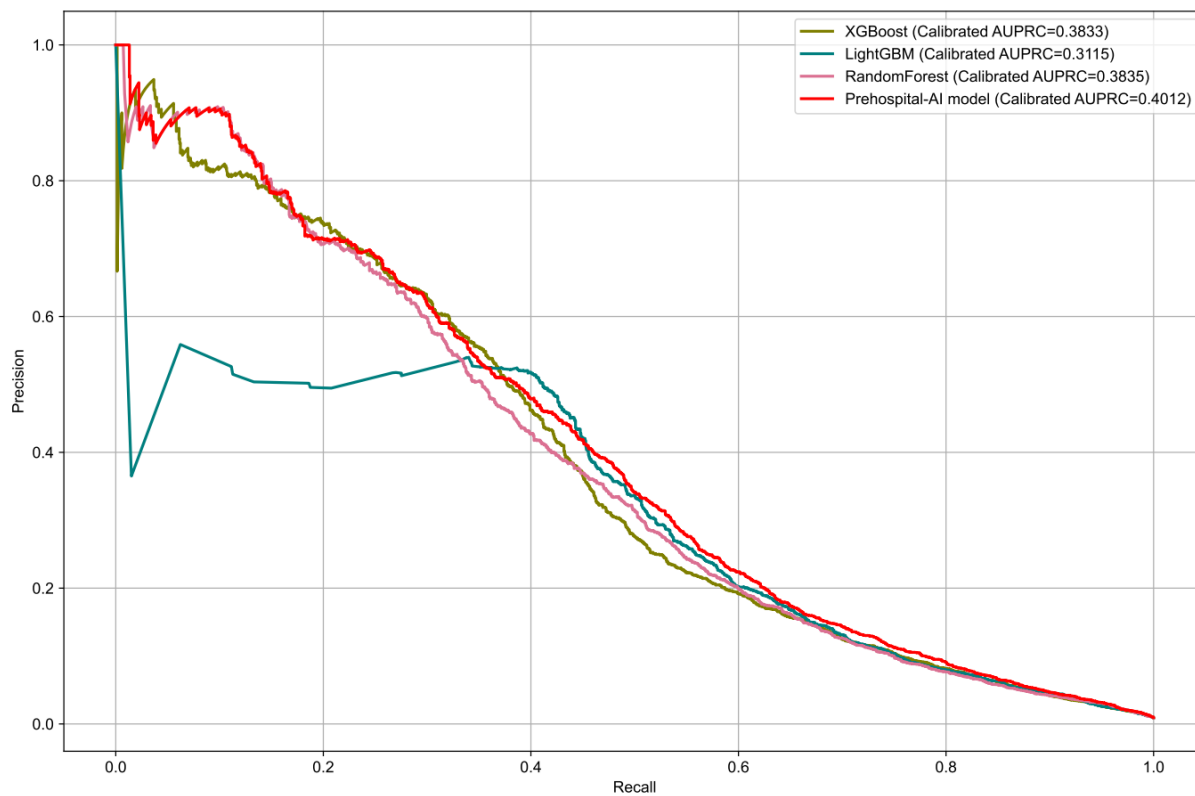


Figure 4. Performance of Calibrated AI Models on the KTDB Test Dataset: AUROC (A) and AUPRC

The original Figures 3 and 4 have been renumbered as Figures 5 and 7, respectively.

We revised in the method section as follows:

Original Text:

“Shock Index Comparison

To further assess the predictive capability of the Prehospital-AI model, its performance was compared with the shock index (SI), a widely used physiologic marker for trauma severity. Details on SI characteristics, data selection and exclusion criteria are provided in **Supplementary Methods 4.**”

Revised Text:

“Model Performance Evaluation and Benchmarking Against Shock Index

To comprehensively assess the predictive performance of the Prehospital-AI model, we compared it with the shock index (SI), a widely used physiologic marker for evaluating trauma severity. The comparison employed a broad set of performance metrics, including sensitivity, specificity, precision, adjusted precision, Lift, accuracy, balanced accuracy, area under the receiver operating characteristic curve (AUROC), area under the precision-recall curve (AUPRC), adjusted precision-recall gain (APG), and normalized AUPRC (nAUPRC). These metrics were chosen to capture various aspects of model discrimination, calibration, and performance under class imbalance. Detailed descriptions of the SI calculation, inclusion and exclusion criteria, and statistical methods are provided in **Supplementary Methods 4.**”

We added relevant references.

Moreover, we added in the Supplementary Methods 4 as follows:

“Lift Metric

In addition to conventional evaluation metrics such as sensitivity, specificity, and the area under the receiver operating characteristic curve (AUROC), we also employed the *Lift* metric to assess the relative improvement of the model compared with random selection. Lift is defined as the ratio between the observed precision (positive predictive value) within a given subset of predictions and the overall prevalence of the positive class in the dataset.

$$Lift = \frac{(TP/(TP + FP))}{(TP + FN)/(TP + TN + FP + FN)}$$

A Lift value of 1 indicates that the model performs no better than random chance, whereas values greater than 1 indicate that the model is more effective in concentrating true positives within the top-ranked predictions.

This metric is particularly informative in highly imbalanced clinical datasets, as it quantifies how many times more likely the model is to identify true positive cases compared with random allocation. Lift charts have been widely adopted in data mining, medical AI, and marketing analytics for model performance evaluation.

This metric is particularly informative in highly imbalanced clinical datasets, because it normalizes the model’s precision against the baseline prevalence of the positive class. In other words, Lift highlights how effectively the model can enrich rare positive cases compared with random selection, thereby providing a more intuitive measure of performance when positive outcomes are scarce.

Adjusted Precision Metric

In addition to conventional evaluation metrics such as sensitivity, specificity, and the area under the receiver operating characteristic curve (AUROC), we employed the **Adjusted Precision** metric to

account for the prevalence of the positive class. Adjusted Precision, also referred to as prevalence-adjusted PPV, measures how much better the model’s precision is compared with random prediction, normalized to a 0–1 scale.

$$Adj.Precision = \frac{(TP/(TP + FP)) - ((TP + FN)/(TP + TN + FP + FN))}{1 - ((TP + FN)/(TP + TN + FP + FN))}$$

A value of 0 indicates performance equivalent to random selection, while a value of 1 represents perfect prediction. Unlike conventional precision, this metric enables fair comparison across datasets with different prevalences, making it particularly informative in highly imbalanced clinical settings.

Adjusted Precision–Recall Gain Metric

We also employed the Adjusted precision–recall Gain (APG) metric to evaluate model performance based on the precision–recall (PR) framework. APG builds on the concept of PR Gain, which measures the improvement of precision over the baseline prevalence while accounting for recall.

$$APG = \frac{AUPRC - ((TP + FN)/(TP + TN + FP + FN))}{1 - ((TP + FN)/(TP + TN + FP + FN))}$$

By integrating PR Gain across all thresholds, APG provides a normalized summary of how effectively the model enriches true positives relative to random selection. An APG of 0 indicates random performance, while a value of 1 reflects perfect classification. This metric is particularly advantageous in rare-event detection tasks, as it jointly incorporates prevalence, precision, and recall, offering a more intuitive interpretation of model performance in highly imbalanced clinical datasets.

nAUPRC (Normalized AUPRC) Metric

To further address the challenges posed by class imbalance, we adopted the Normalized Area Under the Precision–Recall Curve (nAUPRC) as an evaluation metric. The conventional AUPRC

summarizes the trade-off between precision and recall across thresholds, but its absolute value is strongly influenced by the prevalence of the positive class, complicating comparisons across datasets. To mitigate this issue, we normalized the AUPRC against the baseline prevalence:

$$nAUPRC = \frac{AUPRC}{((TP + FN)/(TP + TN + FP + FN))}$$

Here, Prevalence represents the expected AUPRC under random prediction. Thus, $nAUPRC = 0$ corresponds to random performance, and $nAUPRC = 1$ indicates perfect classification. By eliminating the dependence on prevalence, $nAUPRC$ provides a more reliable measure for comparing model performance across datasets with varying degrees of class imbalance.”

We also added relevant references in the supplementary methods.

We added the result of precision and AUPRC to original Table 3. We also added the performance of the calibrated model in additional table. We identified the similar performance of calibrated model compared to original best AI model. Revised table 3 was shown as follows:

Table 3. External validation dataset performance of the Prehospital-AI model and comparison with the Shock index (SI)

Datasets		Sensitivity	Specificity	Precision	Adjusted Precision	Lift	Accuracy	Balanced accuracy	AUROC	AUPRC	APG	nAUPRC
KTDB test set	^a Prehospital-AI model	0.8583	0.8677	0.0576	0.0487	6.1697	0.8676	0.8630	0.9479	0.3927	0.3870	42.0972
	^a Prehospital-AI model calibration	0.8399	0.8828	0.0632	0.0544	6.7746	0.8824	0.8613	0.9479	0.3927	0.3870	42.0972
	^b Prehospital-AI model	0.7953	0.8738	0.0423	0.0356	6.0763	0.8732	0.8345	0.9234	0.1589	0.1530	22.8189
	^b Prehospital-AI model calibration	0.7795	0.8795	0.0434	0.0367	6.2301	0.8788	0.8295	0.9224	0.1628	0.1569	23.3783
	S ^c TH=0.8241	0.5118	0.8165	0.0257	0.0188	3.6839	0.8143	0.6641	0.7117	0.0362	0.0294	5.1997
	I TH=0.9000 ²⁵	0.4488	0.8805	0.0257	0.0188	3.6839	0.8775	0.6646				
CNUH	^a Prehospital-AI model	0.8333	0.8099	0.0455	0.0351	4.2298	0.8101	0.8216	0.9390	0.3346	0.3274	31.1407
	^a Prehospital-AI model calibration	0.8333	0.8177	0.0473	0.0370	4.4033	0.8179	0.8255	0.9381	0.3222	0.3148	29.9819
	^b Prehospital-AI model	0.7778	0.8115	0.0642	0.0487	3.9246	0.8109	0.7946	0.9178	0.1501	0.1360	9.1754
	^b Prehospital-AI model calibration	0.7778	0.8022	0.0614	0.0458	3.7524	0.8018	0.7900	0.9195	0.1416	0.1274	8.6559
	S ^c TH=0.8241	0.6667	0.7431	0.0414	0.0254	2.5287	0.7418	0.7049	0.7766	0.1581	0.1441	9.6595
	I TH=0.9000 ²⁵	0.6667	0.8096	0.0550	0.0393	3.3639	0.8073	0.7381				
CHH	^a Prehospital-AI model	0.9167	0.8550	0.0320	0.0269	6.1529	0.8553	0.8858	0.9489	0.4268	0.4238	82.1293

	^a Prehospital-AI model calibration	0.9167	0.8594	0.0329	0.0279	6.3371	0.8597	0.8880	0.9447	0.4266	0.4236	82.0843
	^b Prehospital-AI model	1.0000	0.9028	0.0160	0.0144	10.1360	0.9029	0.9514	0.9980	0.4167	0.4157	263.9583
	^b Prehospital-AI model calibration	1.0000	0.9036	0.0161	0.0146	10.2177	0.9037	0.9518	0.9976	0.3929	0.3919	248.8750
	S ^c TH=0.8241	0.5000	0.8775	0.0064	0.0048	4.0609	0.8769	0.6887	0.9156	0.0180	0.0165	11.4188
	I TH=0.9000 ²⁵	0.5000	0.9209	0.0099	0.0083	6.2723	0.9203	0.7105				
CBUH	^a Prehospital-AI model	0.8077	0.8817	0.1243	0.1061	6.1031	0.8802	0.8447	0.9300	0.6988	0.6926	34.3236
	^a Prehospital-AI model calibration	0.8077	0.8993	0.1429	0.1250	7.0165	0.8974	0.8535	0.9246	0.7169	0.7110	35.2113
	^b Prehospital-AI model	0.7500	0.8318	0.0619	0.0480	4.2448	0.8306	0.7909	0.8794	0.3652	0.3558	25.0624
	^b Prehospital-AI model calibration	0.7500	0.8540	0.0706	0.0568	4.8441	0.8525	0.8020	0.8759	0.4143	0.4057	28.4325
	S ^c TH=0.8241	0.5000	0.7486	0.0286	0.0142	1.9607	0.7450	0.6243	0.6580	0.1580	0.1455	10.8397
	I TH=0.9000 ²⁵	0.5000	0.7985	0.0354	0.0211	2.4292	0.7942	0.6493				
GUGC	^a Prehospital-AI model	0.9310	0.8077	0.0438	0.0347	4.6733	0.8088	0.8694	0.9562	0.5723	0.5683	61.1197
	^a Prehospital-AI model calibration	0.8966	0.8110	0.0429	0.0339	4.5819	0.8118	0.8538	0.9557	0.5804	0.5765	61.9869
	^b Prehospital-AI model	0.9091	0.8558	0.0490	0.0412	6.0428	0.8562	0.8824	0.9674	0.4205	0.4158	51.8377
	^b Prehospital-AI model calibration	0.9091	0.8632	0.0515	0.0438	6.3543	0.8636	0.8861	0.9667	0.4453	0.4408	54.8980
	S ^c TH=0.8241	0.5455	0.7985	0.0217	0.0137	2.6702	0.7965	0.6720	0.7043	0.0320	0.0241	3.9502
	I TH=0.9000 ²⁵	0.4545	0.8587	0.0256	0.0177	3.1608	0.8555	0.6566				

WH	^a Prehospital-AI model	0.7500	0.7367	0.0032	0.0021	2.8427	0.7367	0.7434	0.8646	0.0347	0.0336	30.9965
	^a Prehospital-AI model calibration	0.7500	0.7728	0.0037	0.0026	3.2926	0.7728	0.7614	0.8946	0.0639	0.0628	57.1266
	^b Prehospital-AI model	-	0.8265	-	-	-	0.8265	0.8265	-	-	-	-
	^b Prehospital-AI model calibration	-	0.8505	-	-	-	0.8505	0.8505	-	-	-	-
	^c TH=0.8241	-	0.7180	-	-	-	0.7180	0.7180	-	-	-	-
	TH=0.9000 ²⁵	-	0.7842	-	-	-	0.7842	0.7842	-	-	-	-
^a Prehospital-AI model: Performance on the entire external validation dataset. ^b Prehospital-AI model: Performance on the subset of the external validation data where SI could be computed for direct comparison. ^c TH=0.8241: Optimal threshold derived from the Youden Index during 5-fold cross-validation.												

We have added a new table presenting the performance metrics of the calibrated model. As shown, the calibrated model demonstrated comparable discrimination performance to the uncalibrated model, while substantially improving calibration metrics. Furthermore, since conventional metrics alone may not fully capture the model’s ability to detect rare events in highly imbalanced clinical settings, we additionally introduced the lift metric. Lift quantifies how effectively the model concentrates positive cases compared with random selection, thereby providing an intuitive and easily interpretable measure of clinical utility in rare-event prediction. As shown in Revised Table 3, the Lift values of our AI model consistently exceeded those of the Shock Index across all datasets, further supporting the robustness and clinical applicability of our approach.

We revised table 2 as follows:

Table 2. Performance comparison of machine learning models and shock index in 5-fold cross-validation

Model	Sensitivity mean (SD)	Specificity mean (SD)	Precision mean (SD)	Adjusted Precision mean (SD)	Lift mean (SD)	Accuracy mean (SD)	Balanced accuracy mean (SD)	AUROC ^a , mean (SD)	AUPRC mean (SD)	APG mean (SD)	nAUPRC mean (SD)
^b XGBoost	0.8696 (0.0165)	0.8531 (0.0076)	0.0540 (0.0021)	0.0451 (0.0021)	5.7814 (0.2184)	0.8562 (0.0074)	0.8628 (0.0056)	0.9371 (0.0059)	0.3881 (0.0326)	0.3823 (0.0329)	41.5470 (3.5028)
Calibrated XGBoost	0.8591	0.8714	0.0593	0.0504	6.3440	0.8713	0.8653	0.9368	0.3833	0.3774	41.0256
^c LightGBM	0.8644(0.0152)	0.8599 (0.0035)	0.0550 (0.0009)	0.0461 (0.0009)	5.8862 (0.0975)	0.8599 (0.0034)	0.8621 (0.0065)	0.9376 (0.0051)	0.3146 (0.0316)	0.3081 (0.0319)	33.6756 (3.4033)
Calibrated LightGBM	0.8519	0.8798	0.0626	0.0538	6.7055	0.8795	0.8658	0.9374	0.3115	0.3050	33.3487
^d Adaboost	0.8735 (0.0106)	0.8393 (0.0050)	0.0488 (0.0016)	0.0398 (0.0016)	5.2245 (0.1749)	0.8396 (0.0049)	0.8564 (0.0046)	0.9359 (0.0062)	0.4249 (0.0407)	0.4195 (0.0411)	45.4900 (4.3855)
Calibrated Adaboost	0.8132	0.9059	0.0754	0.0667	8.0682	0.9051	0.8596	0.9346	0.4169	0.4114	44.6229
^e RF	0.8670 (0.0182)	0.8491 (0.0081)	0.0515 (0.0024)	0.0426 (0.0024)	5.5145 (0.2512)	0.8493 (0.0080)	0.8581 (0.0082)	0.9374 (0.0071)	0.3861 (0.0325)	0.3803 (0.0328)	41.3358 (3.5070)
Calibrated RF	0.8336	0.8872	0.0651	0.0563	6.9737	0.8867	0.8604	0.9372	0.3835	0.3777	41.0552
^f CatBoost	0.8500 (0.0192)	0.8472 (0.0006)	0.0498 (0.0011)	0.0409 (0.0012)	5.3353 (0.1261)	0.8472 (0.0007)	0.8486 (0.0098)	0.9283 (0.0081)	0.3799 (0.0271)	0.3741 (0.0274)	40.6680 (2.9168)
Calibrated CatBoost	0.8637	0.8370	0.0476	0.0386	5.0933	0.8372	0.8503	0.9280	0.3762	0.3703	40.2671
^g GBM	0.8696 (0.0186)	0.8502 (0.0026)	0.0519 (0.0014)	0.0430 (0.0014)	5.5573 (0.1474)	0.8504 (0.0026)	0.8599 (0.0094)	0.9362 (0.0067)	0.3853 (0.0266)	0.3795 (0.0268)	41.2459 (2.8702)
Calibrated GBM	0.8716	0.8493	0.0517	0.0428	5.5375	0.8496	0.8605	0.9361	0.3832	0.3773	41.0153
XGBoost+ LightGBM	0.8729 (0.0136)	0.8628 (0.0051)	0.0566 (0.0014)	0.0477 (0.0014)	6.0620 (0.1414)	0.8629 (0.0049)	0.8678 (0.0047)	0.9400 (0.0055)	0.3968 (0.0346)	0.3911 (0.0350)	42.4781 (3.7310))
Calibrated XGBoost+ LightGBM	0.8532	0.8836	0.0646	0.0558	6.9199	0.8833	0.8684	0.9398	0.3919	0.3861	41.9478
XGBoost+ RF	0.8729 (0.0186)	0.8605 (0.0057)	0.0558 (0.0017)	0.0469 (0.0017)	5.9684 (0.1792)	0.8606 (0.0055)	0.8667 (0.0079)	0.9428 (0.0059)	0.4019 (0.0356)	0.3962 (0.0359)	43.0197 (3.8347)

Calibrated XGBoost+ RF	0.8644	0.8747	0.0611	0.0522	6.5406	0.8747	0.8696	0.9426	0.3985	0.3928	42.6557
LightGBM +RF	0.8735 (0.0233)	0.8610 (0.0036)	0.0560 (0.0018)	0.0471 (0.0018)	5.9898 (0.1878)	0.8611 (0.0035)	0.8672 (0.0114)	0.9426 (0.0052)	0.4017 (0.0371)	0.3961 (0.0375)	43.0043 (4.0031)
Calibrated LightGBM +RF	0.8696	0.8681	0.0585	0.0496	6.2651	0.8681	0.8688	0.9425	0.3985	0.3928	42.6537
Prehospital- AI model	0.8735 (0.0176)	0.8639 (0.0039)	0.0571 (0.0011)	0.0482 (0.0011)	6.1094 (0.1147)	0.8639 (0.0038)	0.8687 (0.0076)	0.9433 (0.0053)	0.4057 (0.0375)	0.4001 (0.0379)	43.4282 (4.0400)
Calibrated Prehospital- AI model	0.8617	0.8792	0.0630	0.0542	6.7464	0.8790	0.8705	0.9431	0.4012	0.3963	43.0266
S I	^h TH =	0.5154	0.8129	0.0208	0.0133	2.7311	0.6641	0.8106			
	0.8241	(0.0080)	(0.0124)	(0.0015)	(0.0015)	(0.2017)	(0.0088)	(0.0122)			
	TH =								0.6656	0.0455	0.0382
	0.9000	0.4355	0.8759	0.0262	0.0188	3.4438	0.6557	0.8725	(0.0203)	(0.0119)	(0.0120)
	₂₅	(0.0250)	(0.0014)	(0.0019)	(0.0019)	(0.2386)	(0.0128)	(0.0015)			5.9701 (1.5417)

We revised supplementary Table 6 as follows:

Supplementary Table 6. Testing dataset performance of the Prehospital-AI model and comparison with the shock index (SI).

	Sensitivity	Specificity	Precision	Adjusted Precision	Lift	Accuracy	Balanced Accuracy	AUROC	AUPRC	APG	nAUPRC
^a Prehospital-AI model	0.8583	0.8677	0.0576	0.0487	6.1697	0.8676	0.8630	0.9479	0.3927	0.3870	42.0972
^a Prehospital-AI model calibration	0.8399	0.8828	0.0632	0.0544	6.7746	0.8824	0.9479	0.8613	0.3927	0.3870	42.0972
^b Prehospital-AI model	0.7953	0.8738	0.0423	0.0356	6.0763	0.8732	0.8345	0.9234	0.1589	0.1530	22.8189
^b Prehospital-AI model calibration	0.7795	0.8795	0.0434	0.0367	6.2301	0.8788	0.8295	0.9224	0.1628	0.1569	23.3783
SI	^c TH=0.8241	0.5118	0.8165	0.0257	0.0188	3.6839	0.8143	0.7117	0.0362	0.0294	5.1997
	TH=0.9000 ⁸	0.4488	0.8805	0.0257	0.0188	3.6839	0.8775				

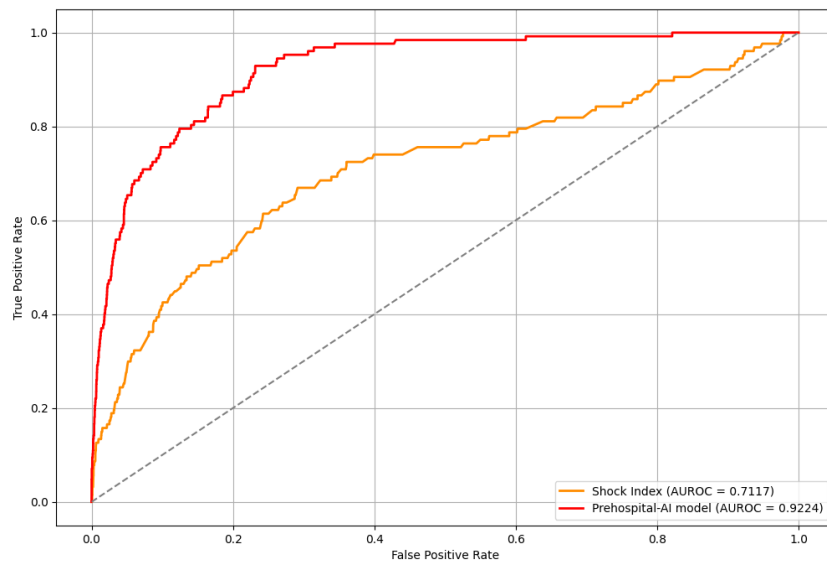
^aPrehospital-AI model: Performance on the entire testing dataset.

^bPrehospital-AI model: Performance on the subset of the testing data where SI could be computed for direct comparison.

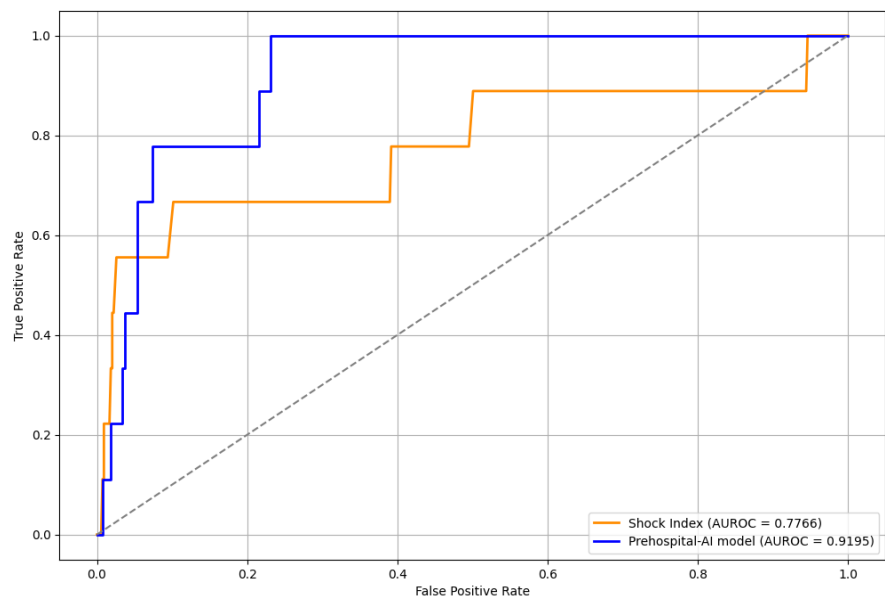
^cTH=0.8241: Optimal threshold derived from the Youden Index during 5-fold cross-validation.

Supplementary Figures 1–5 have been revised to reflect the outputs of the newly calibrated AI model, as detailed below.

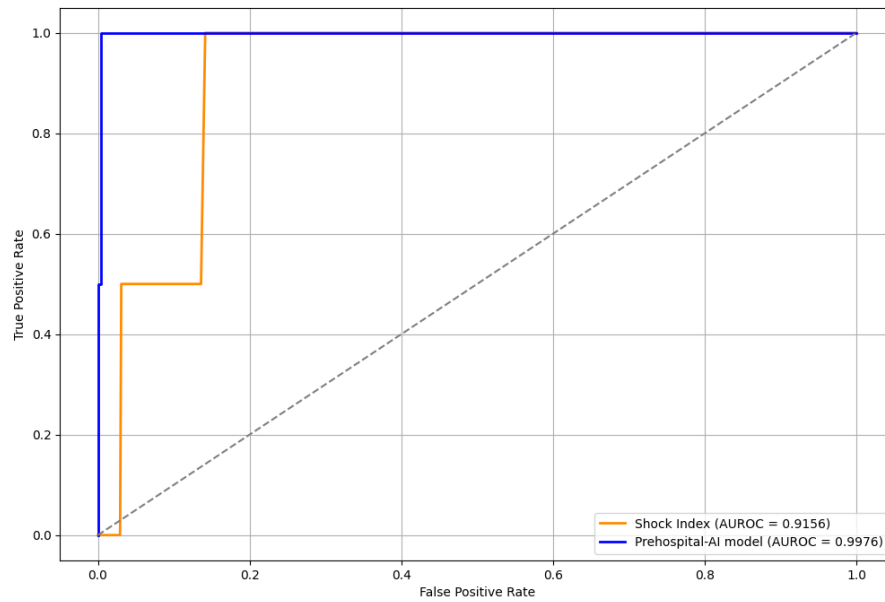
Supplementary Figure 1. Performance comparison between the Shock index and the calibrated Prehospital-AI model on the testing dataset.



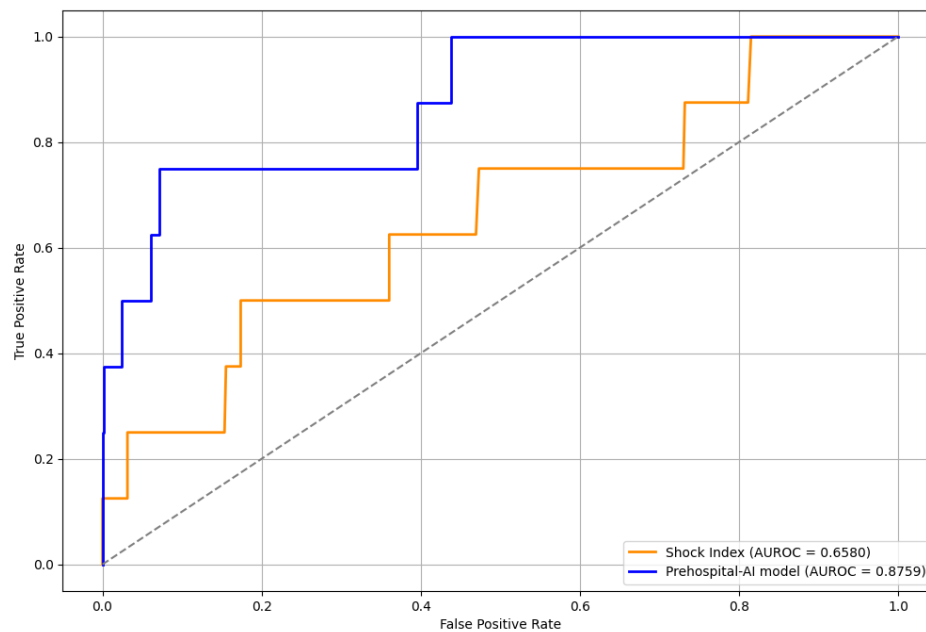
Supplementary Figure 2. Performance comparison of the Shock index and the calibrated Prehospital-AI model using the external validation dataset from Chonnam National University Hospital (CNUH).



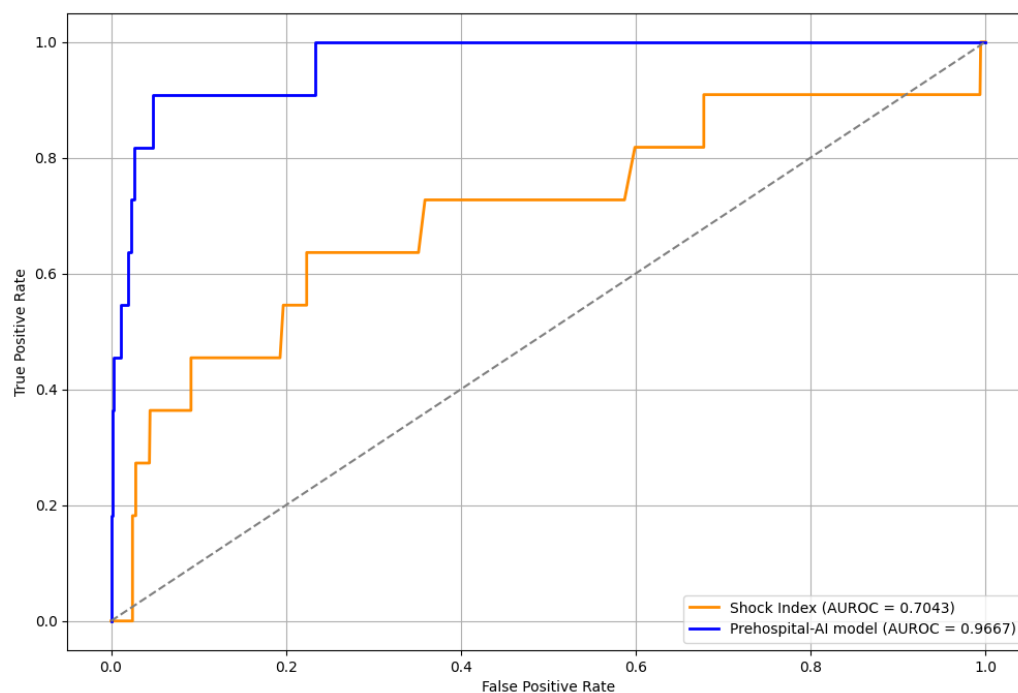
Supplementary Figure 3. Performance comparison of the Shock index and the calibrated Prehospital-AI model using the external validation dataset from Cheju Halla General Hospital (CHH).



Supplementary Figure 4. Performance comparison of the Shock index and the calibrated Prehospital-AI model using the external validation dataset from Chungbuk National University Hospital (CBUH).



Supplementary Figure 5. Performance comparison of the Shock index and the calibrated Prehospital-AI model using the external validation dataset from Gachon University Gil Medical Center (GUGMC).



We revised in the result section as follows:

Original Text:

“To optimize predictive performance, various ensemble models were explored. The Prehospital-AI model, an ensemble of XGBoost, LightGBM, and RF, achieved the highest AUROC (0.9433 ± 0.0053) and balanced accuracy (0.8687 ± 0.0076), surpassing all single models and alternative ensembles. It also demonstrated well-balanced sensitivity (0.8735 ± 0.0176) and specificity (0.8639 ± 0.0039), ensuring reliable classification of trauma patient outcomes.

In contrast, the SI showed significantly lower performance, with AUROC values of 0.6656 ± 0.0203 and balanced accuracy of 0.6641 ± 0.0088 at TH = 0.8241 (determined using the Youden index³⁶). The

performance further declined at the recommended threshold (TH = 0.9000) ³⁷. **Figure 2** in the Supplement compares the ROC curves, highlighting the superior predictive performance of the Prehospital-AI model over individual ML models and the SI.”

Revised Text:

“To optimize predictive performance, various ensemble models were explored. The Prehospital-AI model, an ensemble of XGBoost, LightGBM, and RF, achieved the highest AUROC (0.9433 ± 0.0053) and balanced accuracy (0.8687 ± 0.0076), surpassing all single models and alternative ensembles. It also demonstrated well-balanced sensitivity (0.8735 ± 0.0176) and specificity (0.8639 ± 0.0039), ensuring reliable classification of trauma patient outcomes. **However, the AdaBoost model achieved the highest AUPRC (0.4249 ± 0.0407), compared with 0.4057 ± 0.0375 for the Prehospital-AI model. (Table 2)**

In contrast, the SI showed significantly lower performance, with AUROC values of 0.6656 ± 0.0203 and balanced accuracy of 0.6641 ± 0.0088 at TH = 0.8241 (determined using the Youden index). The performance further declined at the recommended threshold (TH = 0.9000). **Figure 2 compares the AUROC and AUPRC curves, highlighting the superior predictive performance of the Prehospital-AI model over individual ML models and the SI.”**

We revised in the result section as follows:

Original Text:

“Testing Data Results and Comparison with Shock Index

Supplementary Table 6 in the Supplement summarizes the performance of the Prehospital-AI model on the testing dataset, demonstrating strong predictive capability. The model achieved a sensitivity of 0.8583, specificity of 0.8677, accuracy of 0.8676, balanced accuracy of 0.8630, and AUROC of 0.9479, indicating excellent discrimination between survivors and deceased patients. These results are highly

consistent with the 5-fold cross-validation findings in the KTDB dataset, where the model attained a balanced accuracy of 0.8687 and an AUROC of 0.9433. The close agreement between cross-validation and independent testing performance confirms the robustness and generalizability of the model.”

Revised Text:

“Testing Data Results and Comparison with Shock Index

Supplementary Table 6 in the Supplement summarizes the performance of the Prehospital-AI model on the testing dataset, demonstrating strong predictive capability. The **calibrated** model achieved a sensitivity of **0.8399**, specificity of **0.8828**, accuracy of **0.8824**, balanced accuracy of **0.8613**, **AUPRC of 0.3927** and AUROC of 0.9479, indicating excellent discrimination between survivors and deceased patients. These results are highly consistent with the 5-fold cross-validation findings in the KTDB dataset, where the model attained a balanced accuracy of 0.8687 and an AUROC of 0.9433. The close agreement between cross-validation and independent testing performance confirms the robustness and generalizability of the model.”

We revised in the result section as follows:

Original Text:

“External Validation Data Results

Table 3 presents the external validation performance of the Prehospital-AI model, demonstrating its robust predictive capability across diverse clinical settings. In the four South Korean trauma centers, the model consistently exhibited strong predictive performance, with balanced accuracy ranging from 0.8216 to 0.8858. These results closely align with the 5-fold cross-validation outcome (0.8687) and the independent testing dataset performance (0.8630), confirming the model’s generalizability across different trauma centers.

At WH in Australia, the validation posed unique challenges due to a high proportion of missing prehospital data. Of the 21 prehospital variables, only six were fully available, while nine were entirely missing and thus excluded. Furthermore, six key physiological indicators, such as systolic and diastolic blood pressure, pulse rate, respiratory rate, oxygen saturation, and the AVPU scale, were missing in approximately 75% of cases. Despite these substantial data gaps, the Prehospital-AI model maintained a balanced accuracy of 0.7434 and an AUROC of 0.8646, demonstrating its ability to adapt to incomplete datasets while still providing clinically relevant predictions.

To assess its clinical utility, the model was compared against the SI across all external validation sites. The Prehospital-AI model consistently outperformed SI, achieving AUROC values between 0.8794 and 0.9980, whereas SI showed significantly lower AUROCs, ranging from 0.6580 to 0.9156. Notably, in CBUH, SI exhibited particularly poor predictive performance (AUROC: 0.6580, balanced accuracy: 0.6243), reinforcing the limitations of conventional physiologic-based triage tools.“

Revised Text:

“External Validation Data Results

Table 3 presents the external validation performance of the Prehospital-AI model, demonstrating its robust predictive capability across diverse clinical settings. In the four South Korean trauma centers, the **calibrated** model consistently exhibited strong predictive performance, with balanced accuracy ranging from **0.8255** to **0.8880**. These results closely align with the 5-fold cross-validation outcome (0.8687) and the independent testing dataset performance (**0.8613**), confirming the model’s generalizability across different trauma centers.

At WH in Australia, the validation posed unique challenges due to a high proportion of missing prehospital data. Of the 21 prehospital variables, only six were fully available, while nine were entirely missing and thus excluded. Furthermore, six key physiological indicators, such as systolic and diastolic blood pressure, pulse rate, respiratory rate, oxygen saturation, and the AVPU scale, were missing in approximately 75% of cases. Despite these substantial data gaps, the Prehospital-AI model maintained

a balanced accuracy of 0.7614 and an AUROC of 0.8946, demonstrating its ability to adapt to incomplete datasets while still providing clinically relevant predictions.

To assess its clinical utility, the model was compared against the SI across all external validation sites. The Prehospital-AI model consistently outperformed SI, achieving AUROC values between 0.8759 and 0.9976, whereas SI showed significantly lower AUROCs, ranging from 0.6580 to 0.9156. Notably, in CBUH, SI exhibited particularly poor predictive performance (AUROC: 0.6580, balanced accuracy: 0.6243), reinforcing the limitations of conventional physiologic-based triage tools.“

We revised abstract as follows:

Original Text:

“Abstract

Early identification of high-risk trauma patients in the prehospital setting is crucial for optimizing resource allocation and improving survival. Conventional triage tools often rely on limited physiological data and subjective assessments. This study aimed to develop and externally validate a real-time AI model predicting emergency room (ER) mortality in trauma patients using only 21 prehospital features. Model development and internal validation utilized the Korean Trauma Data Bank (KTDB; 204,189 patients). External validation was conducted across four regional trauma centers in South Korea (8,358 patients) and one Level 1 trauma center in Australia (3,578 patients) to assess generalizability. Our Prehospital-AI model, an ensemble of XGBoost, LightGBM, and Random Forest trained on the KTDB using only 21 prehospital variables, demonstrated strong external validation performance across institutions and countries, outperforming the shock index (SI). The Prehospital-AI model achieved an AUROC of 0.923 (sensitivity: 85.3%, specificity: 86.8%) on the testing dataset, compared to the SI's AUROC of 0.712. External validation yielded AUROCs of 0.930-0.956 in South Korean centers and 0.845 in the Australian center. By enabling real-time risk assessment prior to

hospital arrival, this AI-powered model outperforms traditional triage tools and improves trauma system efficiency.”

Revised Text:

“Abstract

Early identification of high-risk trauma patients in the prehospital setting is crucial for optimizing resource allocation and improving survival. Conventional triage tools often rely on limited physiological data and subjective assessments. This study aimed to develop and externally validate a real-time AI model predicting emergency room (ER) mortality in trauma patients using only 21 prehospital features. Model development and internal validation utilized the Korean Trauma Data Bank (KTDB; 204,189 patients). External validation was conducted across four regional trauma centers in South Korea (8,358 patients) and one Level 1 trauma center in Australia (3,578 patients) to assess generalizability. **We applied a post hoc calibration procedure using the beta calibration method.** Our Prehospital-AI model, an ensemble of XGBoost, LightGBM, and Random Forest trained on the KTDB using only 21 prehospital variables, demonstrated strong external validation performance across institutions and countries, outperforming the shock index (SI). The Prehospital-AI model achieved an AUROC of 0.923 (sensitivity: **0.780**, specificity: **0.880**) on the testing dataset, compared to the SI's AUROC of 0.712. External validation yielded AUROCs of **0.925**–0.956 in South Korean centers and **0.895** in the Australian center. By enabling real-time risk assessment prior to hospital arrival, this AI-powered model outperforms traditional triage tools and improves trauma system efficiency.”

Regarding external validation, Table 3 shows high AUROC values at domestic facilities in South Korea but lower AUROC values for the Australian dataset. Although the data collection periods differ, this raises the possibility of overfitting, and the international generalizability of the model remains uncertain.

Response> Thank you for your insightful comment. The Australian dataset presented substantial

heterogeneity in variable distributions and contained a considerable amount of missing data, which presented significant challenges for external validation. Despite these limitations, the model's performance—sensitivity of 0.75, specificity of 0.7367, and AUROC of 0.8646—appears favorable. In the case of the WH dataset, there were only 4 death cases among 3,578 records, resulting in an extremely imbalanced class distribution with a positive prevalence of approximately 0.14%. In such settings where positive cases are exceptionally rare, evaluation metrics that focus on the positive class—such as precision and AUPRC—are inevitably low in absolute terms due to structural limitations. Specifically, precision is defined as the proportion of true positives (TP) among all positive predictions (TP + FP), and when the positive class is exceedingly sparse, even a small number of false positives (FP) can cause a steep decline in the precision score. Our model was designed with an emphasis on securing sensitivity under this highly imbalanced context, and therefore, the trade-off of reduced precision was an anticipated and acceptable outcome. Accordingly, rather than interpreting the model's performance solely based on absolute precision values, we additionally report auxiliary metrics that allow a fairer comparison with the baseline. For example, after applying calibration, the model achieved a lift of 3.2926 and a normalized AUPRC (nAUPRC) of 57.1266 on the WH dataset, representing a substantial improvement over the baseline precision of 0.0014. These results indicate that the model performs well above random guessing and retains meaningful discriminative capacity even under conditions of extreme class rarity. We acknowledge the potential risk of overfitting; however, the results remain promising, particularly in comparison with the study by Yu et al., who reported strong internal validation performance (AUROC 0.938 in Korea) but notable variability across external datasets (AUROC 0.850 in Malaysia, 0.756 in Taiwan, and 0.836 in Vietnam) in their multinational validation study.

We added in the limitation section as follows:

“The performance of external validation using the Australian dataset was lower than that observed in the Korean dataset. A similar trend was reported by Yu et al., who developed and externally validated the GIFT score—an interpretable, machine learning-based triage model derived from multinational prehospital data across Asia. While the model demonstrated strong internal validity in the Korean

dataset (AUROC 0.938), its performance varied across external datasets, with AUROC values of 0.850 in Malaysia, 0.756 in Taiwan, and 0.836 in Vietnam. In line with these findings, our model showed reduced discriminative performance in the Australian cohort, potentially due to a higher proportion of missing values and differences in data characteristics, as reflected by the lower AUROC compared to the Korean dataset.”

We also added the relevant references.

In the response of reviewer’s concern, we revised the conclusion as follows:

Original text:

“Our Prehospital-AI model demonstrated high predictive accuracy and robustness across internal and external validation datasets, confirming its potential as a real-time decision-support tool for trauma care. By relying solely on prehospital variables, it enables early mortality risk assessment without requiring hospital-based scoring systems, bridging a critical gap in trauma triage. The model seamlessly integrates with emergency medical systems, complementing previous AI models for in-hospital and ED mortality prediction. Its real-time application supports AI-driven trauma management across multiple stages, optimizing prehospital decision-making and resource allocation. Future work will focus on developing a multimodal AI system incorporating audio from emergency conversations, wearable device monitoring, and medical imaging. Expanding beyond tabular data will enhance predictive capabilities and improve emergency care workflows, ultimately advancing trauma patient outcomes.”

Revised text:

“Our Prehospital-AI model demonstrated high predictive accuracy and robustness across internal and external validation datasets, confirming its potential as a real-time decision-support tool for trauma care. By relying solely on prehospital variables, it enables early mortality risk assessment without requiring hospital-based scoring systems, bridging a critical gap in trauma triage. The model seamlessly integrates with emergency medical systems, complementing previous AI models for in-hospital and ED mortality

prediction. Its real-time application supports AI-driven trauma management across multiple stages, optimizing prehospital decision-making and resource allocation. However, the potential risk of overfitting—especially in heterogeneous external environments such as the Australian dataset, which contains numerous missing values—highlights the need for further validation using standardized and internationally representative datasets. Such efforts will enhance the model’s generalizability and ensure consistent performance across diverse trauma systems and populations. Future work will focus on developing a multimodal AI system incorporating audio from emergency conversations, wearable device monitoring, and medical imaging. Expanding beyond tabular data will enhance predictive capabilities and improve emergency care workflows, ultimately advancing trauma patient outcomes.”

From a triage perspective, given the reported sensitivity and specificity, and considering the low outcome prevalence, the positive predictive value would be only a few percent, resulting in a high number of false positives. The actual benefit of using this predictive model is unclear. A Decision Curve Analysis would be useful in this context, but such an evaluation was not conducted.

Response> Thank you for your valuable comment. In response to the reviewer’s suggestion, we performed Decision Curve Analysis (DCA) to assess the clinical utility of our model. Consistent with the calibration results, the calibrated model demonstrated superior performance in the DCA. In the KTDB test dataset, the calibrated model showed a higher net benefit across a range of threshold probabilities, particularly in the clinically relevant low-threshold range, indicating improved decision-making potential.

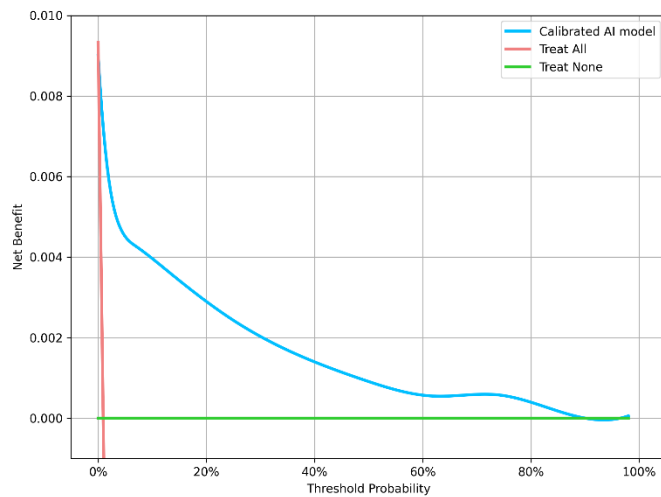


Figure 6 (A). Decision Curve Analysis Comparing Calibrated and Original AI Models in the KTDB Test Dataset

The figure above compares the net benefit of different strategies across a range of threshold probabilities. The blue line represents the calibrated AI model, which demonstrates a consistently higher net benefit than the default strategies of treat all (red) or treat none (green), especially in the low threshold range. This suggests that calibration not only improved probabilistic accuracy but also enhanced potential clinical applicability by optimizing decision-making across plausible thresholds. These findings support the value of model calibration in improving the real-world utility of AI-driven predictions.

We also performed decision curve analysis (DCA) on the external validation set, and the results have been included in the result section and figures as follows:

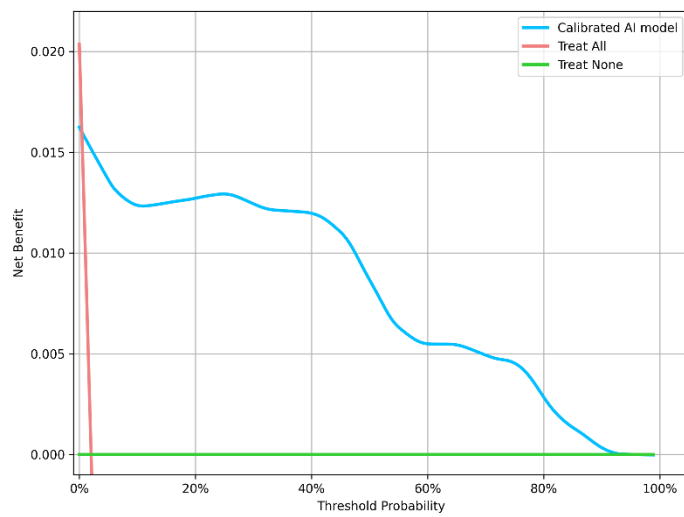


Figure 6 (B). Decision Curve Analysis Comparing Calibrated and Original AI Models in the CBUH Dataset

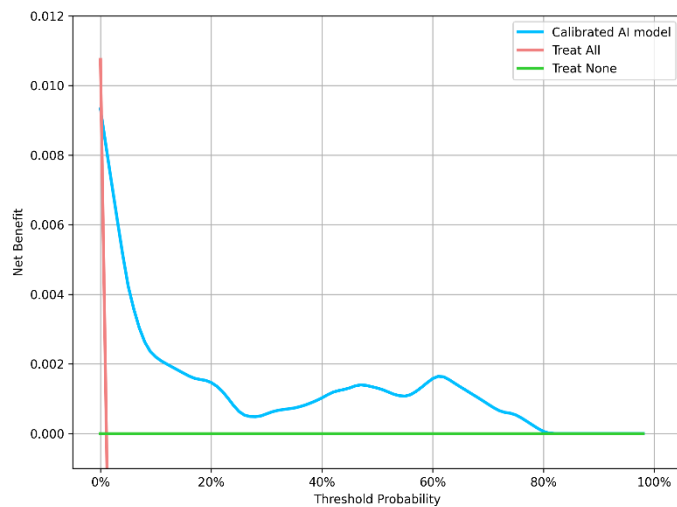


Figure 6 (C). Decision Curve Analysis Comparing Calibrated and Original AI Models in the CNUH Dataset

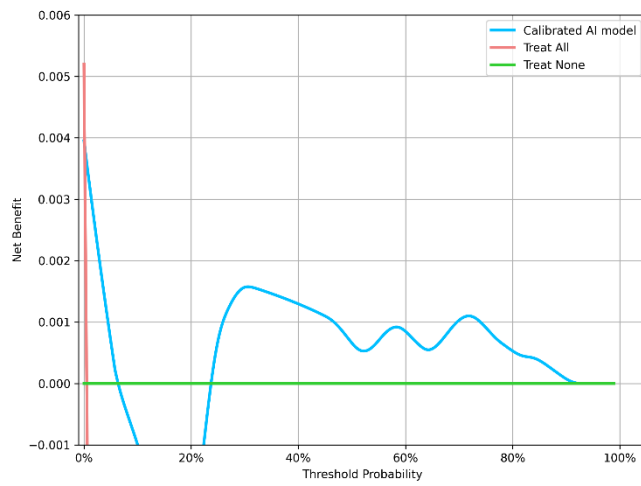


Figure 6 (D). Decision Curve Analysis Comparing Calibrated and Original AI Models in the CHH Dataset

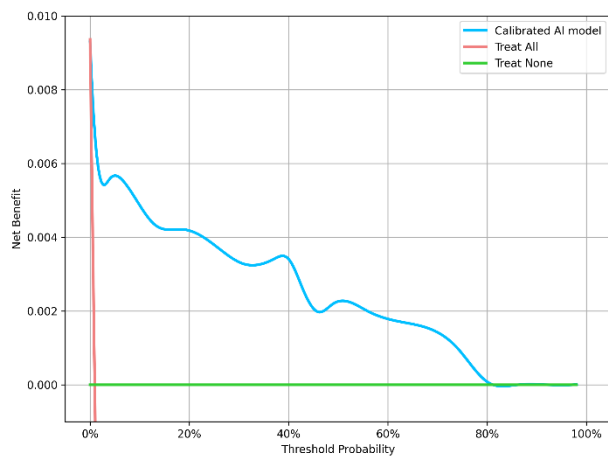


Figure 6 (E). Decision Curve Analysis Comparing Calibrated and Original AI Models in the GUGC Dataset

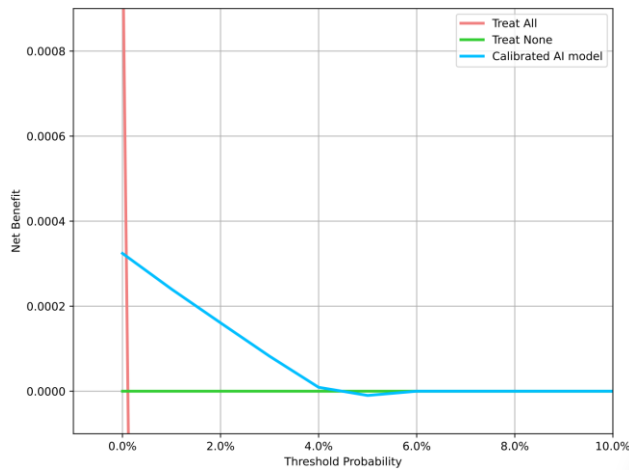
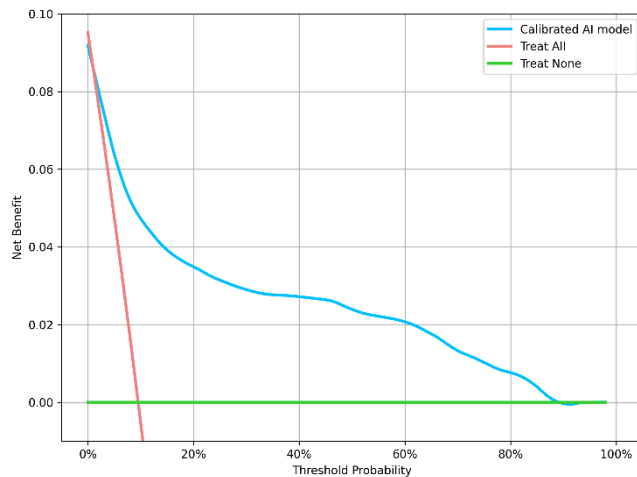


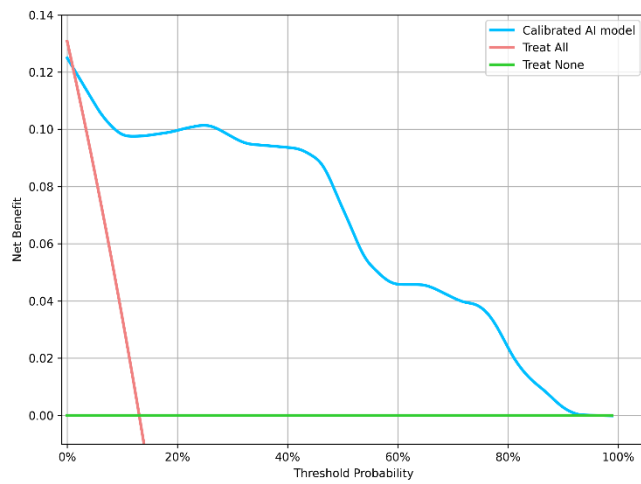
Figure 6 (F). Decision Curve Analysis Comparing Calibrated and Original AI Models in the WH Dataset

This has been added to new Figure 6 in the Results section.

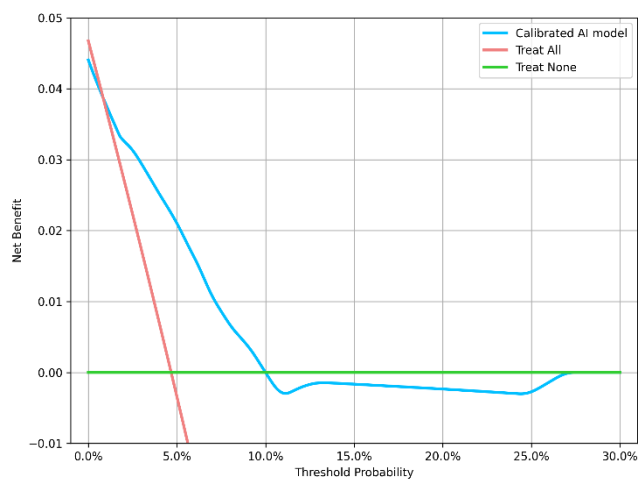
Moreover, we performed DCA specifically in the subgroup of high-risk patients with impaired mental status. This subgroup included patients presenting with coma, semicoma, or drowsiness. The results of the decision curve analysis in the high-risk group are presented as follows:



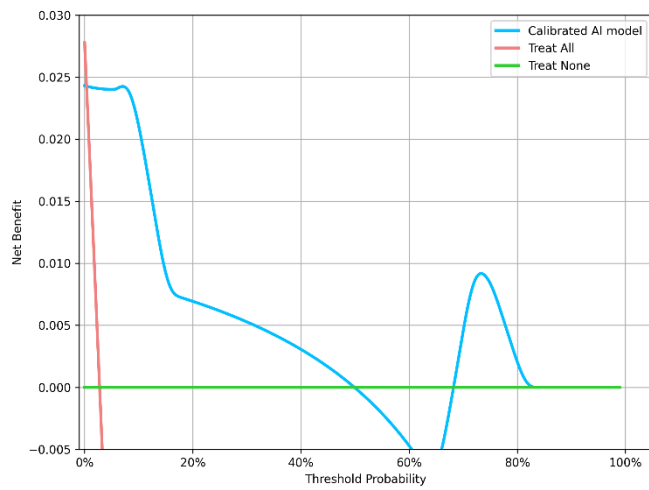
Supplementary Figure 7 (A). Decision Curve Analysis in the KTDB Test Dataset: High Risk Group with Altered Mental Status (N = 7,112)



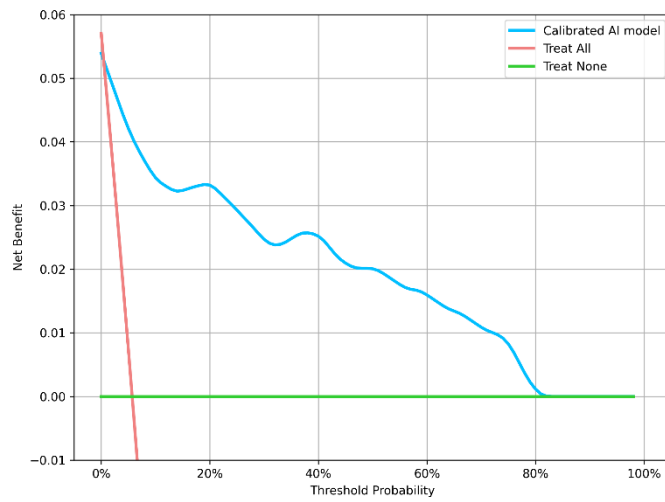
Supplementary Figure 7 (B). Decision Curve Analysis in the CBUH Dataset: High Risk Group with Altered Mental Status (N = 153)



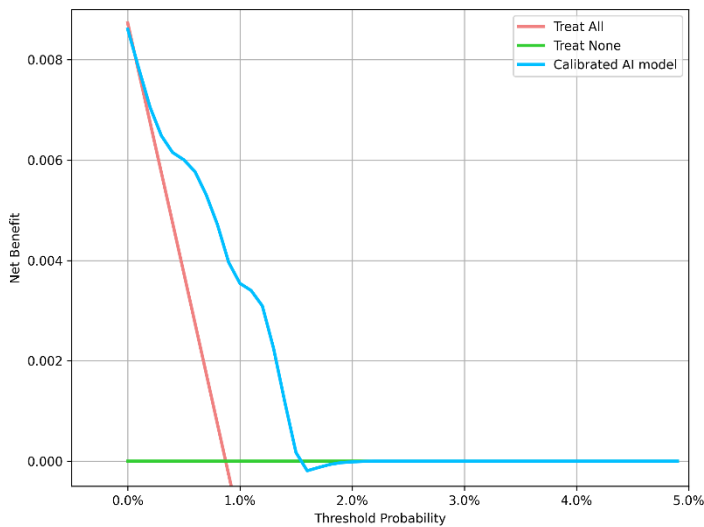
Supplementary Figure 7 (C). Decision Curve Analysis in the CNUH Dataset: High Risk Group with Altered Mental Status (N = 107)



Supplementary Figure 7 (D). Decision Curve Analysis in the CHH Dataset: High Risk Group with Altered Mental Status (N = 108)



Supplementary Figure 7 (E). Decision Curve Analysis in the GUGC Dataset: High Risk Group with Altered Mental Status (N = 298)



Supplementary Figure 7 (F). Decision Curve Analysis in the WH Dataset: High Risk Group with Altered Mental Status (N = 458)

This has been added to Supplementary Figure 7.

DCA of the AI models in the KTDB test dataset is presented in new Figure 5. The blue line represents the calibrated AI model, which consistently demonstrated a higher net benefit than the default strategies of treating all (red) or treating none (green), particularly within the lower threshold range. This suggests that the calibrated AI model may enhance clinical applicability by optimizing decision-making across plausible thresholds. These findings highlight the importance of calibration in improving the real-world utility of AI-driven predictions. Similarly, DCA of the external validation datasets showed that the calibrated AI model provided a higher net benefit, particularly within the 0–1% mortality range, which reflects real-world prevalence. In addition, we performed a subgroup analysis in high-risk patients with impaired mental status, including those presenting with coma, semicoma, or drowsiness.

We added in the method section with relevant reference as follows:

“To further evaluate the clinical utility of the model, we perform decision curve analysis. This method assesses the net benefit of the model across various threshold probabilities, thereby illustrating its

potential value in clinical decision-making. The observation of greater net benefit compared to treat-all or treat-none strategies within relevant thresholds supports its clinical applicability.”

We added in the result section as follows:

“Decision Curve Analysis

Decision curve analysis (DCA) of the AI models in the KTDB test dataset is shown in Figure 5. The blue line represents the calibrated AI model, which consistently demonstrated a higher net benefit than the default strategies of treating all (red) or treating none (green), particularly within the lower threshold range. This indicates that the calibrated AI model may enhance clinical applicability by optimizing decision-making across plausible thresholds. These findings underscore the value of calibration in improving the real-world utility of AI-driven predictions. Similarly, DCA of the external validation datasets showed that the calibrated AI model provided a higher net benefit, particularly within the 0–1% mortality range, which reflects real-world prevalence. Notably, in the high-risk subgroup characterized by altered mental status, the AI model also yielded superior net benefit. (supplementary figure 7).“

We revised the supplementary method as follows:

Original text:

“The models were implemented using Python (version 3.8.19) with the following libraries: NumPy (version 1.23.5), Pandas (version 2.0.3), Matplotlib (version 3.7.5), and Scikit-learn (version 1.3.0). Statistical analyses were conducted using R (version 4.4.0) (R Foundation for Statistical Computing, Vienna, Austria). Categorical variables were compared using the χ^2 test or Fisher exact test, as appropriate, with statistical significance set at a two-tailed P value of < 0.05 .”

Revised text:

“The models were implemented using Python (version 3.8.19) with the following libraries: NumPy (version 1.23.5), Pandas (version 2.0.3), **Betacal (version 1.1.0)**, Matplotlib (version 3.7.5), and Scikit-learn (version 1.3.0). Statistical analyses were conducted using R (version 4.4.0) (R Foundation for Statistical Computing, Vienna, Austria), **employing the following packages: tidyverse, MoonBook, and dcurves**. Categorical variables were compared using the χ^2 test or Fisher exact test, as appropriate, with statistical significance set at a two-tailed P value of < 0.05 .”

Reviewer #1 (Remarks on code availability):

The code for model training and testing is adequate. However, since the data has not been shared and the trained model is not provided, it is difficult to verify the reproducibility of the predictions.

Response> Thank you for your valuable comments. In response to the reviewers’ suggestions, we have added the anonymized sample dataset to the repository.

Reviewer #2 (Remarks to the Author):

The manuscript by Oh et al. presents a real-time AI model designed to predict emergency room (ER) mortality among pre-hospital trauma patients. The model was trained on retrospective data from the Korean Trauma Data Bank (KTDB) and externally validated using retrospective datasets from four regional trauma centers in South Korea and one Level 1 trauma center in Australia. The model demonstrated strong performance in both internal and external validations, outperforming the Shock Index. While the study offers a promising approach for early mortality prediction in trauma care, several major concerns need to be addressed:

1. Data Representation in Table 1: The data presentation in Table 1 is unclear, particularly for variables with categories labeled as "uncheckable" or "unchecked." For instance, the SpO₂ (%) feature includes 3.0% as uncheckable and 51.2% as unchecked, alongside subcategories of 0.2%, 1.1%, 2.7%, 6.1%, and 86.9% for different value ranges. These proportions do not add up to 100%, leading to confusion about how the data is categorized and interpreted.

Response> Thank you for your critical comment. We have revised the figures in Table 1 accordingly. In the KTDB dataset, the code “-1” denotes “uncheckable” values, which are assigned when the patient is under 15 years of age, refuses vital sign measurement, or when measurement is not feasible due to unknown reasons. The code “-9” indicates “unchecked” values, typically applied when patients are not directly transferred to the trauma center via ambulance or air transportation. Additionally, some variables in the KTDB dataset have entirely missing values; these cases were treated as missing data in our analysis. We added footnote in the Table 1, as follows:

“Uncheckable: patients <15 years old, refusal to measure vital signs, or measurement infeasible due to unknown reasons; Unchecked: patients not directly transferred to the trauma center via ambulance or air transportation; Missing data: entirely absent in the original dataset.”

We revised the supplementary method 2 as follows:

Original text:

“Prehospital vital signs were categorized into ordinal groups, with additional specific codes for cases where measurements were unavailable or not meaningful. A value of 1 (uncheckable) indicated that the vital sign could not be assessed due to severe injury or an unresponsive state. A value of 2 (unchecked) was assigned when a measurement was not performed, such as in patients younger than 14 years. A value of 3 (zero) was recorded when a measurement was attempted but was too low to be physiologically viable. Missing values were handled using multiple imputations, replacing them with the mean value from the training dataset to minimize data loss and maintain model integrity.”

Revised text:

“Prehospital vital signs were categorized into ordinal groups, with additional specific codes for cases where measurements were unavailable or not meaningful. In the KTDB dataset, the code “-1” denotes “uncheckable” values, which are assigned when the patient is under 15 years of age, refuses vital sign measurement, or when measurement is not feasible due to unknown reasons. The code “-9” indicates “unchecked” values, typically applied when patients are not directly transferred to the trauma center via ambulance or air transportation. Additionally, some variables in the KTDB dataset have entirely missing values; these cases were treated as missing data in our analysis. Missing values were handled using mean imputations, replacing them with the mean value from the training dataset to minimize data loss and maintain model integrity.”

We revised the Table 1 as follows:

Table 1. Comparison of patient characteristics and prehospital variables across KTDB (training/testing dataset), Korean trauma centers (4 external validation datasets), and the Australian trauma center (1 external validation dataset)

Variables	Training and test sets (KTDB)		External validation sets (5 datasets)		p
	KTDB (test set) (N=40838)	KTDB (train set) (N=163351)	Four Korean trauma centers (CNUH, CHGH, CBUH and GUGC) (N=8358)	Australian trauma center (WH) (N=3578)	
ED mortality	381 (0.9%)	1526 (0.9%)	85 (1.0%)	4 (0.1%)	<0.001
In-hospital mortality	1609 (3.9%)	6317 (3.9%)	364 (4.4%)	130 (3.5%)	0.085
ISS	10.5 ± 9.2	10.5 ± 9.2	10.5 ± 9.1	8.7 ± 7.8	<0.001
Time of injury occurrence					
Daytime (8 am–5 pm)	19464 (47.7%)	77547 (47.5%)	3742 (44.8%)	0 (0%)	<0.001
Nighttime (5 pm–8 am)	20941 (51.2%)	84034 (51.4%)	3884 (46.5%)	0 (0%)	<0.001
Missing	433 (1.1%)	1770 (1.1%)	732 (8.8%)	3578 (100.0%)	<0.001
Age	-	-	56.7 ± 22.4	51.4 ± 22.4	<0.001
Age (categorized by 5-year-old unit), year					
< 1	125 (0.3%)	537 (0.3%)	36 (0.4%)	0 (0.0%)	<0.001
1-4	489 (1.2%)	1945 (1.2%)	81 (1.0%)	0 (0.0%)	<0.001

5-9	912 (2.2%)	3500 (2.1%)	136 (1.6%)	0 (0.0%)
10-14	809 (2.0%)	3289 (2.0%)	187 (2.2%)	0 (0.0%)
15-19	1227 (3.0%)	4768 (2.9%)	215 (2.6%)	107 (3.0%)
20-24	1446 (3.5%)	5997 (3.7%)	270 (3.2%)	307 (8.6%)
25-29	1643 (4.0%)	6407 (3.9%)	320 (3.8%)	289 (8.1%)
30-34	1488 (3.6%)	5793 (3.5%)	356 (4.3%)	275 (7.7%)
35-39	1755 (4.3%)	7066 (4.3%)	290 (3.5%)	246 (6.9%)
40-44	1949 (4.8%)	7853 (4.8%)	394 (4.7%)	247 (6.9%)
45-49	2502 (6.1%)	10340 (6.3%)	414 (5.0%)	233 (6.5%)
50-54	3246 (7.9%)	13327 (8.2%)	625 (7.5%)	228 (6.4%)
55-59	4104 (10.0%)	16010 (9.8%)	714 (8.5%)	220 (6.1%)
60-64	4057 (9.9%)	16447 (10.1%)	814 (9.7%)	236 (6.6%)
65-69	3400 (8.3%)	13339 (8.2%)	755 (9.0%)	248 (6.9%)
70-74	3066 (7.5%)	12197 (7.5%)	677 (8.1%)	216 (6.0%)
75-79	3394 (8.3%)	13549 (8.3%)	686 (8.2%)	227 (6.3%)
80-84	2906 (7.1%)	11941 (7.3%)	706 (8.4%)	194 (5.4%)
85-89	1698 (4.2%)	6513 (4.0%)	470 (5.6%)	161 (4.5%)
90-94	511 (1.3%)	2077 (1.3%)	169 (2.0%)	112 (3.1%)
100-104	93 (0.2%)	399 (0.2%)	36 (0.4%)	26 (0.7%)
105-109	16 (0.0%)	47 (0.0%)	3 (0.0%)	5 (0.1%)
110-114	2 (0.0%)	7 (0.0%)	3 (0.0%)	0 (0.0%)
115-119	0 (0.0%)	2 (0.0%)	1 (0.0%)	0 (0.0%)

> 120	0 (0.0%)	1 (0.0%)	0 (0.0%)	1 (0.0%)	
Male sex	26634 (65.2%)	105668 (64.7%)	5116 (61.2%)	2375 (66.4%)	<0.001
Intentionality					
Accidental, unintentional	37779 (92.5%)	150880 (92.5%)	7280 (87.1%)	2444 (68.3%)	<0.001
Self-harm, suicide	760 (1.9%)	3000 (1.8%)	207 (2.5%)	124 (3.5%)	<0.001
violence, assault	768 (1.9%)	3017 (1.8%)	102 (1.2%)	356 (9.9%)	<0.001
other specified	292 (0.7%)	1311 (0.8%)	6 (0.1%)	3 (0.1%)	<0.001
unspecified	585 (1.4%)	2494 (1.5%)	81 (1.0%)	651 (18.2%)	<0.001
Missing	654 (1.6%)	2649 (1.6%)	682 (8.2%)	0 (0.0%)	<0.001
Injury mechanism					
car accident	5190 (12.7%)	20905 (12.9%)	760 (9.1%)	539 (15.1%)	<0.001
bike accident	1465 (3.6%)	5602 (3.4%)	233 (2.8%)	112 (3.1%)	0.002
motorcycle accident	3211 (7.9%)	12989 (8.0%)	567 (6.8%)	363 (10.1%)	<0.001
other traffic accidents	4256 (10.4%)	16384 (10.0%)	698 (8.4%)	28 (0.8%)	<0.001
unspecified traffic accident	20 (0.0%)	65 (0.0%)	0 (0.0%)	96 (2.7%)	<0.001
fall	7603 (18.7%)	30701 (18.8%)	1505 (18.0%)	1660 (46.4%)	<0.001
slipped	10187 (24.9%)	40564 (24.9%)	2145 (25.7%)	0 (0.0%)	<0.001
struck	3619 (8.9%)	14441 (8.8%)	618 (7.4%)	404 (11.3%)	<0.001
firearm, cut, or pierced	2180 (5.3%)	8814 (5.4%)	527 (6.3%)	247 (6.9%)	<0.001
machine	1084 (2.7%)	4560 (2.8%)	233 (2.8%)	45 (1.3%)	<0.001

fire, flames, or heat	124 (0.3%)	515 (0.3%)	51 (0.6%)	6 (0.2%)	<0.001
drowning	15 (0.0%)	56 (0.0%)	0 (0.0%)	2 (0.1%)	0.334
choking, hanging	11 (0.0%)	64 (0.0%)	1 (0.0%)	28 (0.8%)	<0.001
others	1105 (2.7%)	4440 (2.7%)	318 (3.8%)	33 (0.9%)	<0.001
unknown	768 (1.9%)	3251 (2.0%)	131 (1.6%)	15 (0.4%)	<0.001
missing data	0 (0.0%)	0 (0.0%)	571 (6.8%)	0 (0.0%)	<0.001
Type of injury					
blunt	37799 (92.6%)	150902 (92.4%)	7069 (84.6%)	3256 (91.0%)	<0.001
penetrating	2502 (6.1%)	10247 (6.3%)	632 (7.6%)	320 (8.9%)	<0.001
fire	311 (0.8%)	1329 (0.8%)	52 (0.6%)	0 (0.0%)	<0.001
others	142 (0.3%)	556 (0.3%)	5 (0.1%)	0 (0.0%)	<0.001
unknown	84 (0.2%)	317 (0.2%)	29 (0.3%)	1 (0.0%)	0.002
missing data	0 (0.0%)	0 (0.0%)	571 (6.8%)	1 (0.0%)	<0.001
Job-related					
yes	5949 (14.6%)	23630 (14.5%)	1294 (15.5%)	0 (0%)	<0.001
no	30154 (73.8%)	120680 (73.9%)	5877 (70.3%)	0 (0%)	<0.001
unknown	4735 (11.6%)	19041 (11.6%)	1055 (12.6%)	0 (0%)	<0.001
missing	0 (0.0%)	0 (0.0%)	132 (1.6%)	3578 (100.0%)	<0.001
Accident location					
home	8077 (19.8%)	32263 (19.8%)	2013 (24.1%)	480 (13.4%)	<0.001

residential area	1246 (3.1%)	4907 (3.0%)	0 (0.0%)	12 (0.3%)	<0.001
accommodation	171 (0.4%)	685 (0.4%)	46 (0.6%)	11 (0.3%)	0.199
company/office	115 (0.3%)	506 (0.3%)	4 (0.0%)	6 (0.2%)	<0.001
factory/workplace	1814 (4.4%)	7283 (4.5%)	393 (4.7%)	7 (0.2%)	<0.001
construction site	1223 (3.0%)	4975 (3.0%)	164 (2.0%)	43 (1.2%)	<0.001
school/education facility	429 (1.1%)	1822 (1.1%)	100 (1.2%)	3 (0.1%)	<0.001
general road	10533 (25.8%)	42003 (25.8%)	1616 (19.3%)	1032 (28.8%)	<0.001
highway	594 (1.5%)	2207 (1.4%)	96 (1.1%)	0 (0.0%)	<0.001
medical institution	688 (1.7%)	2658 (1.6%)	114 (1.4%)	113 (3.2%)	<0.001
mountain/rice field/farm/livestock pen	2020 (4.9%)	7874 (4.8%)	380 (4.5%)	56 (1.6%)	<0.001
community facility	1197 (2.9%)	4737 (2.9%)	0 (0.0%)	102 (2.9%)	<0.001
stream/river/reservoir/sea	504 (1.2%)	1949 (1.2%)	112 (1.3%)	11 (0.3%)	<0.001
transportation facility	270 (0.7%)	1137 (0.7%)	56 (0.7%)	0 (0.0%)	<0.001
bicycle path/sidewalk	1120 (2.7%)	4567 (2.8%)	316 (3.8%)	74 (2.1%)	<0.001
military area	92 (0.2%)	350 (0.2%)	6 (0.1%)	0 (0.0%)	0.001
cultural/public facility	243 (0.6%)	893 (0.5%)	65 (0.8%)	7 (0.2%)	0.002
sports facility	245 (0.6%)	1023 (0.6%)	178 (2.1%)	160 (4.5%)	<0.001
commercial facility	549 (1.3%)	2078 (1.3%)	344 (4.1%)	55 (1.5%)	<0.001
others	5856 (14.4%)	23554 (14.4%)	399 (4.8%)	237 (6.6%)	<0.001
unknown	3851 (9.4%)	15879 (9.7%)	1824 (21.8%)	1169 (32.7%)	<0.001
missing data	1 (0.0%)	1 (0.0%)	132 (1.6%)	0 (0.0%)	<0.001

Hospital arrival route

direct visit	26127 (64.0%)	103987 (63.7%)	5671 (67.9%)	0 (0%)	<0.001
transferred from another facility	14249 (34.9%)	57625 (35.2%)	2592 (31.0%)	0 (0%)	<0.001
referred from outpatient	459 (1.1%)	1727 (1.1%)	94 (1.1%)	0 (0%)	<0.001
others	1 (0.0%)	10 (0.0%)	1 (0.0%)	0 (0%)	0.653
unknown	2 (0.0%)	2 (0.0%)	0 (0.0%)	0 (0%)	0.454
missing data	0 (0.0%)	0 (0.0%)	0 (0.0%)	3578 (100.0%)	<0.001

Mode of arrival

emergency medical service ambulance	19789 (48.5%)	78819 (48.3%)	4298 (51.4%)	0 (0%)	<0.001
medical institution ambulance	2454 (6.0%)	10009 (6.1%)	664 (7.9%)	0 (0%)	<0.001
other ambulances	7182 (17.6%)	28890 (17.7%)	1098 (13.1%)	0 (0%)	<0.001
police or other public vehicles	27 (0.1%)	107 (0.1%)	4 (0.0%)	0 (0%)	0.424
air transport	805 (2.0%)	3226 (2.0%)	61 (0.7%)	0 (0%)	<0.001
other vehicles	10460 (25.5%)	41830 (25.6%)	2120 (25.4%)	0 (0%)	<0.001
on foot	95 (0.2%)	408 (0.2%)	110 (1.3%)	0 (0%)	<0.001
others	24 (0.1%)	59 (0.0%)	3 (0.0%)	0 (0%)	0.124
unknown	2 (0.0%)	3 (0.0%)	0 (0.0%)	0 (0%)	0.655
missing data	0 (0.0%)	0 (0.0%)	0 (0.0%)	3578 (100.0%)	<0.001

Insurance type

health insurance (regional, workplace)	26355 (64.6%)	105592 (64.7%)	6047 (72.3%)	0 (0%)	<0.001
automobile insurance	9391 (23.0%)	37256 (22.8%)	1273 (15.2%)	0 (0%)	<0.001
industrial accident compensation insurance	1207 (3.0%)	4938 (3.0%)	299 (3.6%)	0 (0%)	<0.001
private insurance	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0%)	>0.999
medical aid type 1	1841 (4.5%)	7308 (4.5%)	482 (5.8%)	0 (0%)	<0.001
medical aid type 2	302 (0.7%)	1185 (0.7%)	78 (0.9%)	0 (0%)	<0.001
general	1570 (3.8%)	6476 (4.0%)	166 (2.0%)	0 (0%)	<0.001
others	157 (0.4%)	545 (0.3%)	13 (0.2%)	0 (0%)	<0.001
unknown	15 (0.0%)	51 (0.0%)	0 (0.0%)	0 (0%)	0.239
missing data	0 (0.0%)	0 (0.0%)	0 (0.0%)	3578 (100.0%)	<0.001
Pre-hospital cardiac arrest status					
yes	343 (0.8%)	1299 (0.8%)	79 (0.9%)	0 (0%)	<0.001
no	39735 (97.3%)	159059 (97.4%)	7704 (92.2%)	0 (0%)	<0.001
unknown	760 (1.9%)	2993 (1.8%)	443 (5.3%)	0 (0%)	<0.001
missing data	0 (0.0%)	0 (0.0%)	132 (1.6%)	3578 (100.0%)	<0.001
Pre-hospital notification status					
yes	15401 (37.7%)	61431 (37.6%)	4500 (53.8%)	0 (0%)	<0.001
no	10507 (25.7%)	42570 (26.1%)	464 (5.6%)	0 (0%)	<0.001
not applicable	7308 (17.9%)	29436 (18.0%)	1592 (19.0%)	0 (0%)	<0.001

unknown	7622 (18.7%)	29914 (18.3%)	1670 (20.0%)	0 (0%)	<0.001
missing data	0 (0.0%)	0 (0.0%)	132 (1.6%)	3578 (100.0%)	<0.001
Initial AVPU scale					
Alert	16347 (40.0%)	64363 (39.4%)	3171 (37.9%)	455 (12.7%)	<0.001
Verbal response (drowsy)	292 (0.7%)	1291 (0.8%)	266 (3.2%)	306 (8.6%)	<0.001
Painful response (semicoma)	376 (0.9%)	1557 (1.0%)	265 (3.2%)	45 (1.3%)	<0.001
Unresponsive (coma)	718 (1.8%)	2878 (1.8%)	200 (2.4%)	107 (3.0%)	<0.001
unknown	2203 (5.4%)	9135 (5.6%)	139 (1.7%)	0 (0.0%)	<0.001
Missing data	20902 (51.2%)	84127 (51.4%)	4317 (51.7%)	2665 (74.5%)	<0.001
Prehospital SBP, mmHg					
Uncheckable	1673 (4.1%)	6799 (4.2%)	491 (5.9%)	0 (0.0%)	<0.001
Unchecked	20902 (51.2%)	84127 (51.5%)	3772 (45.1%)	0 (0.0%)	<0.001
Missing data	0 (0.0%)	0 (0.0%)	365 (4.4%)	2686 (75.1%)	<0.001
0	6 (0.0%)	30 (0.0%)	7 (0.1%)	13 (0.4%)	<0.001
1–49	2 (0.0%)	18 (0.0%)	1 (0.0%)	3 (0.1%)	<0.001
50–75	328 (0.8%)	1314 (0.8%)	78 (0.9%)	34 (1.0%)	0.404
76–89	436 (1.1%)	1731 (1.1%)	100 (1.2%)	23 (0.6%)	0.042
≥89	17491 (42.8%)	69332 (42.4%)	3544 (42.4%)	819 (22.9%)	<0.001
Prehospital DBP, mmHg					

Uncheckable	1914 (4.7%)	7727 (4.7%)	499 (6.0%)	0 (0.0%)	<0.001
Unchecked	20902 (51.2%)	84127 (51.6%)	3772 (45.1%)	0 (0.0%)	<0.001
Missing data	0 (0.0%)	0 (0.0%)	365 (4.4%)	2753 (76.9%)	<0.001
0	28 (0.1%)	78 (0.0%)	7 (0.1%)	7 (0.2%)	0.001
1–29	13 (0.0%)	70 (0.0%)	5 (0.1%)	4 (0.1%)	0.141
30–45	294 (0.7%)	1273 (0.8%)	74 (0.9%)	24 (0.7%)	0.3
46–59	549 (1.3%)	2207 (1.4%)	145 (1.7%)	69 (1.9%)	0.001
≥59	17138 (42.0%)	67869 (41.5%)	3491 (41.8%)	721 (20.2%)	<0.001
Prehospital PR, beats per min					
Uncheckable	1119 (2.7%)	4511 (2.8%)	361 (4.3%)	0 (0.0%)	<0.001
Unchecked	20902 (51.2%)	84127 (51.5%)	3772 (45.1%)	0 (0.0%)	<0.001
Missing data	0 (0.0%)	0 (0.0%)	365 (4.4%)	2678 (74.8%)	<0.001
0	34 (0.1%)	150 (0.1%)	8 (0.1%)	10 (0.3%)	0.005
1–29	20 (0.0%)	79 (0.0%)	3 (0.0%)	2 (0.1%)	0.961
30–59	450 (1.1%)	1767 (1.1%)	113 (1.4%)	25 (0.7%)	0.01
60–100	15344 (37.6%)	60928 (37.3%)	3011 (36.0%)	590 (16.5%)	<0.001
101–119	2037 (5.0%)	7958 (4.9%)	472 (5.6%)	147 (4.1%)	0.001
≥ 119	932 (2.3%)	3831 (2.3%)	253 (3.0%)	126 (3.5%)	<0.001

Prehospital RR, breaths per min

Uncheckable	1156 (2.8%)	4788 (2.9%)	359 (4.3%)	0 (0.0%)	<0.001
Unchecked	20902 (51.2%)	84127 (51.5%)	3772 (45.1%)	0 (0.0%)	<0.001
Missing data	0 (0.0%)	0 (0.0%)	365 (4.4%)	2684 (75.0%)	<0.001
0	36 (0.1%)	170 (0.1%)	8 (0.1%)	12 (0.3%)	<0.001
1–5	8 (0.0%)	18 (0.0%)	3 (0.0%)	4 (0.1%)	<0.001
6–9	25 (0.1%)	77 (0.0%)	4 (0.0%)	10 (0.3%)	<0.001
10–29	18515 (45.3%)	73376 (44.9%)	3785 (45.3%)	796 (22.2%)	<0.001
≥ 29	196 (0.5%)	795 (0.6%)	62 (0.7%)	72 (2.0%)	<0.001

Prehospital BT (°C)

Uncheckable	1493 (3.7%)	6072 (3.7%)	404 (4.8%)	0 (0.0%)	<0.001
Unchecked	20902 (51.2%)	84127 (51.5%)	3772 (45.1%)	0 (0.0%)	<0.001
Missing data	0 (0.0%)	0 (0.0%)	365 (4.4%)	3578 (100.0%)	<0.001
0	32 (0.1%)	169 (0.1%)	2 (0.0%)	0 (0%)	0.015
0–24.0	2 (0.0%)	2 (0.0%)	0 (0.0%)	0 (0%)	0.454
24.1–28.0	1 (0.0%)	1 (0.0%)	0 (0.0%)	0 (0%)	0.727
28.1–32.0	4 (0.0%)	6 (0.0%)	0 (0.0%)	0 (0%)	0.356

32.1–35.0	316 (0.8%)	1196 (0.7%)	88 (1.1%)	0 (0%)	<0.001
35.1–37.8	17666 (43.3%)	70039 (42.9%)	3619 (43.3%)	0 (0%)	<0.001
≥ 37.8	422 (0.9%)	1739 (1.1%)	108 (1.3%)	0 (0%)	<0.001
SpO2 (%)					
Uncheckable	1225 (3.0%)	5017 (3.1%)	399 (4.8%)	0 (0.0%)	<0.001
Unchecked	20902 (51.2%)	84127 (51.5%)	3772 (45.1%)	0 (0.0%)	<0.001
Missing data	0 (0.0%)	0 (0.0%)	0 (0.0%)	2702 (75.5%)	<0.001
0	80 (0.2%)	301 (0.2%)	368 (4.4%)	8 (0.2%)	<0.001
1–80	437 (1.1%)	1737 (1.1%)	74 (0.9%)	21 (0.6%)	0.019
81–90	1101 (2.7%)	4629 (2.8%)	256 (3.1%)	63 (1.8%)	<0.001
91–95	2504 (6.1%)	9616 (5.9%)	566 (6.8%)	174 (4.9%)	<0.001
≥ 95	14589 (35.7%)	57924 (35.4%)	2923 (35.0%)	610 (17.0%)	<0.001
Uncheckable vital sign including SBP, DBP, PR, RR, or SpO2	2387 (5.8%)	9661 (5.9%)	556 (6.7%)	0 (0.0%)	<0.001
Protective use					
yes	6508 (15.9%)	26604 (16.3%)	1037 (12.4%)	0 (0%)	<0.001
no	13503 (33.1%)	53539 (32.8%)	3373 (40.4%)	0 (0%)	<0.001
unknown	20827 (51.0%)	83208 (50.9%)	3948 (47.2%)	3578 (100.0%)	<0.001

ED, emergency department; KTDB, Korean Trauma Data Bank; SBP, systolic blood pressure; DBP, diastolic blood pressure, PR, pulse rate; RR,

respiratory rate, BT, body temperature; ISS, injury severity score; AVPU, Alert/Verbal/Painful/Unresponsive; **Uncheckable:** patients <15 years old, refusal to measure vital signs, or measurement infeasible due to unknown reasons; **Unchecked:** patients not directly transferred to the trauma center via ambulance or air transportation; **Missing data:** entirely absent in the original dataset

2. Model Interpretability: While SHAP values were used to identify the importance of input features in predicting mortality, the manuscript lacks a discussion on model interpretability. The authors should elaborate on how these key features relate to trauma mortality and provide clinical insights or implications for practice.

Response> Thank you for your valuable comment. According to reviewer's comment, we added in the discussion section as follows:

“In our study, SHAP summary analysis identified several key predictors of emergency department (ED) mortality. Higher oxygen saturation ($\text{SpO}_2 > 95\%$), alert mental status, daytime injury occurrence, and transport by non-ambulance vehicles were consistently associated with a lower risk of mortality. Similarly, the use of protective equipment and injury mechanisms such as slip-down incidents were linked to improved survival. In contrast, advanced age, decreased consciousness, nighttime injuries, lack of protective equipment, and high-energy mechanisms such as collisions were strongly associated with increased mortality risk. Interestingly, missing or unknown prearrival reports and unrecorded accident locations were also associated with higher model-predicted mortality. Although these variables may not directly reflect patient physiology, they may indicate operational patterns of emergency responders in high-stress or time-critical scenarios. Taken together, these findings support the clinical plausibility of the model's predictions and enhance interpretability by aligning with established trauma risk factors. Furthermore, unlike conventional studies with limited sample sizes, AI-based approaches leveraging large-scale data may offer deeper insights into latent patterns within trauma care systems.”

3. Comparative Evaluation: The model's performance is only compared against the Shock Index. However, as noted in the Introduction, other existing models (referenced in citations 6, 7, 8, and 13) also address similar tasks. A thorough comparison and discussion of how the proposed model performs relative to these existing approaches is necessary to better contextualize its contribution.

Response> Thank you for your valuable comment. We revised one paragraph with relevant references

in the discussion section as follows:

Original text:

“Currently, conventional triage tools are used in the field, including the US-adopted Korean triage guidelines and Australia’s major trauma transport protocol (T1). However, these systems are complex, often relying on subjective judgment and incomplete variables. In our Australian dataset, all patients met T1 activation criteria, yet only 0.1% died in the ED, underscoring the need for more precise triage tools for severe trauma cases.”

Revised text:

“Currently, conventional triage tools are employed in the field, including the U.S.-adopted Korean triage guidelines and Australia’s Major Trauma Triage Tool (T1). The 2021 U.S. field triage guideline update by Newgard et al. emphasized the need for evidence-based, simplified criteria using likelihood ratios and AUROC to inform decision-making, aiming to reduce undertriage while improving field usability. However, these systems often remain complex, relying on subjective judgment and frequently incomplete variables. In our Australian dataset, all patients met the T1 activation criteria, yet only 0.1% experienced ED mortality, underscoring the need for more precise triage tools in severe trauma cases.

Digital innovations have shown promise in overcoming these limitations. In a large-scale quality improvement study, Lokerman et al. demonstrated that implementing a mobile app-based triage tool in the Netherlands significantly reduced undertriage without increasing overtriage, using the Injury Severity Score (ISS) as the reference standard. Choi et al. introduced a multitask deep learning model that quantifies injury severity in real time using in-hospital data, achieving AUROCs of 0.82 for mortality prediction and 0.83 for ISS classification in external validation. However, its reliance on post-admission variables limits its applicability in prehospital settings. Yu et al. developed and externally validated the GIFT score, an interpretable machine learning-based triage model derived from multinational prehospital data across Asia. The model showed strong performance in internal validation (AUROC 0.938 in Korea), but performance varied across external datasets (AUROC 0.850 in Malaysia,

0.756 in Taiwan, and 0.836 in Vietnam). Similarly, our external validation in the Australian dataset yielded lower AUROC values compared to Korean data. Our Prehospital-AI model is uniquely designed to predict mortality in real time using only 21 distinct prehospital variables. It incorporates ensemble machine learning techniques and outperforms traditional tools such as the shock index, maintaining robust predictive accuracy even in external datasets with substantial missing data. Notably, our model was externally validated using an Australian dataset with markedly different trauma epidemiology and system characteristics compared to Korea, demonstrating its potential generalizability across heterogeneous healthcare settings. As a scalable and practical solution, Prehospital-AI offers the potential to enhance the accuracy and efficiency of trauma triage both domestically and internationally.”

4. Reproducibility and Code Availability: Although the authors have shared scripts for data preprocessing, model training, and validation via a GitHub repository, important resources are missing. Specifically, detailed scripts and documentation of the data preprocessing steps, access to raw or sample datasets, and the trained model itself (and the web-based AI system for mobile use) are not provided. These omissions hinder the reproducibility of the study's results.

Response> We thank the reviewer for highlighting the importance of end-to-end reproducibility.

Accordingly, we have made the following resources publicly available in our GitHub repository (<https://github.com/ohnaeun111/Prehospital-AI>) and via our web demo:

1. Detailed Preprocessing Code & Documentation

- The `preprocess_trauma_data` function in `Preprocess Trauma Data.py` implements every preprocessing step in a single pipeline. Detailed usage examples and command-line flags are provided in `README.md` under Trauma Data Preprocessing. The steps include:

1. Timestamp Parsing

- Convert raw injury and symptom timestamps to datetime vectors
(`pd.to_datetime(..., errors='coerce')`).

2. Time-of-Day Categorization

- After extracting the time component (`.dt.hour`) from the injury timestamp, each record is categorized into one of three groups according to the hour: "day" if $09:00 \leq \text{time} < 17:00$, "night" otherwise, "missing_data" if the time is null. Corresponding binary flags (`injury_time_day`, `injury_time_night`, `injury_time_missing`) are then generated for each category.

3. Age Binning

- Group `age_years` into 5-year bins (e.g., 0–4 → group 1, 5–9 → group 2, ..., 125+ → group 26) via a custom `bin_age` routine.

4. Categorical One-Hot Encoding

- Map and one-hot encode each key categorical field—such as gender, intentionality, injury type, injury mechanism, and hospital visit route—using predefined dictionaries.

5. Vital-Sign Range Categorization

- Define clinically meaningful intervals for SBP, DBP, pulse, respiration, body temperature, and SpO_2 and create binary flags for each range (e.g., `sbp_50_75`, `pulse_60_100`).

6. “Uncheckable”/“Unchecked” Flags

- For each vital sign, generate additional binary indicators whenever the raw value is coded as `−1` (“uncheckable”) or `−9` (“unchecked”).

To run this entire pipeline, users simply invoke the Python script with input and output arguments on the command line.

2. Sample Dataset

We provide a fully preprocessed, de-identified sample dataset (Sample_dataset.csv), allowing users to clone the repository and reproduce the entire feature engineering pipeline by simply running the following command

3. Training & Evaluation Scripts

We have uploaded train.py (including SMOTE configuration and 5-fold cross-validation loops) and evaluate.py (for computing AUROC, AUPRC etc.)

4. Trained Model Weights & Inference Code

In the Prehospital-AI model/ folder, we have included five fold-specific model files (Prehospital_AI_model_fold1.pkl through Prehospital_AI_model_fold5.pkl) and the Model_Loader.py script demonstrating how to load these weights and perform ensemble predictions.

5. Web-Based Demonstration

A live deployment of the Prehospital-AI system is available at

http://ai-wm.khu.ac.kr/Prehospital_Trauma_Mortality/

With these resources, any researcher can fully reproduce our study—from raw data preprocessing through model training, evaluation, prediction, and web deployment—without additional correspondence.

Reviewer #2 (Remarks on code availability):

Although the authors have shared scripts for data preprocessing, model training, and validation via a GitHub repository, important resources are missing. Specifically, detailed scripts and documentation of the data preprocessing steps, access to raw or sample datasets, and the trained model itself (and the web-based AI system for mobile use) are not provided. These omissions hinder the reproducibility of the

study's results.

Response> We appreciate your insightful comment. As it is identical to the previous one, we refer to our response provided above.

Response Letter

(2nd revision)

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We would like to express our sincere gratitude to the reviewers for their thoughtful and constructive feedback on our manuscript. We carefully reviewed each comment and have revised the manuscript accordingly.

We deeply appreciate the reviewers' efforts in improving the clarity, rigor, and impact of our work. Below, we provide a point-by-point response to each comment.

We have carefully revised the document to make the reviewer's comments and our responses clearly distinguishable. As requested, our responses are marked in blue text, and changes in the manuscript are highlighted in red for clarity. In addition, page numbers have been added to facilitate easy cross-referencing.

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

In this revision, the evaluation using AUROC and precision, calibration, and decision curve analysis that were previously pointed out have been added, improving the interpretability of the results in the context of imbalanced data. The external validation using data from Korea appears to be generally robust, and the DCA results also suggest clinical usefulness.

On the other hand, for the Australian dataset, although the AUROC appears favorable at 0.895, the AUPRC is quite low at 0.064. The DCA shows a marginally higher net benefit, but only within a very narrow range of thresholds, and the overall net benefit is considerably lower compared to that based on Korean data. This is likely due to the influence of missing values and differences in emergency medical systems between countries. Given that some features used in the prediction model inherently reflect country-specific system characteristics, this limitation is unavoidable and also indicates that the generalizability of the model to other countries is limited.

Therefore, as mentioned at the beginning of the Discussion, it would be inaccurate to state that the model performed well even for the Australian data. A more appropriate interpretation would be that while the external validation using Korean data yielded robust results, the model's usefulness for the Australian dataset remains limited due to differences in healthcare systems and data collection processes.

Response> Thank you for your thoughtful and constructive comments. We fully agree with your assessment regarding the limited generalizability of our model to the Australian dataset. In response, we have carefully revised the abstract, discussion, and conclusion to ensure that the interpretation aligns with the reviewer's observation.

Specifically, we have clarified that while the external validation using Korean datasets demonstrated robust and consistent results, the model's performance for the Australian dataset was comparatively lower. We now explicitly attribute this to substantial missing prehospital variables and differences in healthcare systems, documentation practices, and prehospital workflows. Accordingly, the statement that the model "performed well" for the Australian dataset has been removed and replaced with a more accurate description emphasizing the limitation of generalizability.

To further reflect this revision, the abstract was refined within the 200-word limit and now includes a concise statement emphasizing the need for future multinational studies to evaluate model generalizability across diverse trauma systems. Similarly, the first paragraph of the Discussion has been updated to clarify the likely causes of reduced performance in the Australian cohort, and the Conclusion has been refined to underscore the importance of international validation using standardized datasets.

The detailed revisions are presented below.

We revised the Abstract as follows:

Original Text:

“Early identification of high-risk trauma patients in the prehospital setting is crucial for optimizing resource allocation and improving survival. Conventional triage tools often rely on limited physiological data and subjective assessments. This study aimed to develop and externally validate a real-time AI model predicting emergency room (ER) mortality in trauma patients using only 21 prehospital features. Model development and internal validation utilized the Korean Trauma Data Bank (KTDB; 204,189 patients). External validation was conducted across four regional trauma centers in South Korea (8,358 patients) and one Level 1 trauma center in Australia (3,578 patients) to assess generalizability. We applied a post hoc calibration procedure using the beta calibration method. Our Prehospital-AI model, an ensemble of XGBoost, LightGBM, and random forest trained on the KTDB using only 21 prehospital variables, found strong external validation performance across institutions and countries, outperforming the shock index (SI). The Prehospital-AI model achieved an AUROC of 0.923 (sensitivity: 0.780, specificity: 0.880) on the testing dataset, compared to the SI's AUROC of 0.712. External validation yielded AUROCs of 0.925–0.956 in South Korean centers and 0.895 in the Australian center. By enabling real-time risk assessment prior to hospital arrival, this AI-powered model outperforms traditional triage tools and improves trauma system efficiency.”

Revised Text:

“Early identification of high-risk trauma patients in the prehospital setting is crucial for optimizing resource allocation and improving survival. ~~Conventional triage tools often rely on limited physiological data and subjective assessments.~~ This study aimed to develop and externally validate a real-time AI model predicting emergency room (ER) mortality in trauma patients using only 21 prehospital features. Model development and internal validation utilized the Korean Trauma Data Bank (KTDB; 204,189 patients). External validation was conducted across four regional trauma centers in South Korea (8,358 patients) and one Level 1 trauma center in Australia (3,578 patients) to assess generalizability. ~~We applied a post hoc calibration procedure using the beta calibration method.~~ Our Prehospital-AI model, an ensemble of XGBoost, LightGBM, and random forest trained on the KTDB using only 21 prehospital variables, found strong external validation performance across institutions and countries, outperforming the shock index (SI). The Prehospital-AI model achieved an AUROC of 0.923 (sensitivity: 0.780, specificity: 0.880) on the testing dataset, compared to the SI's AUROC of

0.712. External validation yielded AUROCs of 0.925–0.956 in South Korean centers and 0.895 in the Australian center. By enabling real-time risk assessment prior to hospital arrival, this AI-powered model outperforms traditional triage tools and improves trauma system efficiency. **Nonetheless, additional multinational studies are warranted to further evaluate the model’s generalizability across diverse trauma care systems”**

We revised the first paragraph in the discussion section as follows:

Original Text:

“We developed and validated the Prehospital-AI model, demonstrating its ability to accurately predict ED mortality at the prehospital stage. The model achieved high performance, with a sensitivity of 0.8735, specificity of 0.8639, accuracy of 0.8639, balanced accuracy of 0.8687, and an AUROC of 0.9433. Notably, it performed well in external validation, including the Australian dataset, which had substantial missing prehospital variables. The Prehospital-AI model consistently outperformed the conventional SI across all external validation datasets. To enhance accessibility, we deployed a web application with a user-friendly interface optimized for mobile use.”

Revised Text:

“We developed and validated the Prehospital-AI model, demonstrating its ability to accurately predict ED mortality at the prehospital stage. The model achieved high performance, with a sensitivity of 0.8735, specificity of 0.8639, accuracy of 0.8639, balanced accuracy of 0.8687, and an AUROC of 0.9433. **External validation results were similarly robust, with the exception of the Australian dataset, where performance was lower—likely due to substantial missing prehospital variables. This limitation may reflect differences in healthcare systems, data documentation practices, and prehospital workflows. The high proportion of missing data in the Australian cohort presented a significant challenge for model generalizability. Despite this,** the Prehospital-AI model consistently outperformed the conventional SI across all external validation datasets. To enhance accessibility, we deployed a web application with a user-friendly interface optimized for mobile use.”

We revised the conclusion as follows:

Original Text:

“Our Prehospital-AI model demonstrated high predictive accuracy and robustness across internal and external validation datasets, confirming its potential as a real-time decision-support tool for trauma care.

By relying solely on prehospital variables, it enables early mortality risk assessment without requiring hospital-based scoring systems, bridging a critical gap in trauma triage. The model seamlessly integrates with emergency medical systems, complementing previous AI models for in-hospital and ED mortality prediction. Its real-time application supports AI-driven trauma management across multiple stages, optimizing prehospital decision-making and resource allocation. However, the potential risk of overfitting—especially in heterogeneous external environments such as the Australian dataset, which contains numerous missing values—highlights the need for further validation using standardized and internationally representative datasets. Such efforts will enhance the model’s generalizability and ensure consistent performance across diverse trauma systems and populations. Future work will focus on developing a multimodal AI system incorporating audio from emergency conversations, wearable device monitoring, and medical imaging. Expanding beyond tabular data will enhance predictive capabilities and improve emergency care workflows, ultimately advancing trauma patient outcomes.”

Revised Text:

“Our Prehospital-AI model demonstrated high predictive accuracy and robustness across internal and external validation datasets, confirming its potential as a real-time decision-support tool for trauma care. By relying solely on prehospital variables, it enables early mortality risk assessment without requiring hospital-based scoring systems, bridging a critical gap in trauma triage. The model seamlessly integrates with emergency medical systems, complementing previous AI models for in-hospital and ED mortality prediction. Its real-time application supports AI-driven trauma management across multiple stages, optimizing prehospital decision-making and resource allocation. However, the potential risk of overfitting—especially in heterogeneous external environments such as the Australian dataset, which contains numerous missing values **and showed comparatively lower performance**—highlights the need for further validation using standardized and internationally representative datasets. Such efforts will enhance the model’s generalizability and ensure consistent performance across diverse trauma systems and populations. Future work will focus on developing a multimodal AI system incorporating audio from emergency conversations, wearable device monitoring, and medical imaging. Expanding beyond tabular data will enhance predictive capabilities and improve emergency care workflows, ultimately advancing trauma patient outcomes.”

Reviewer #2 (Remarks to the Author):

The authors have addressed most of the concerns I raised in the previous round. However, I am unable to access the GitHub repository and the web server, which are intended to provide the code and model for application.

Response> Thank you for your valuable comment. It appears that the link may not have worked due to a trailing period at the end of the URL. For accurate access, please use the following address without punctuation: https://github.com/ohnaeun111/Pre_hospital_mortality

To avoid confusion, we have revised it as follows:

Original Text:

“Code Availability Statement

The statistical and analytic code is available at https://github.com/ohnaeun111/Pre_hospital_mortality.”

Revised Text:

“Code Availability Statement

The statistical and analytic code is available (https://github.com/ohnaeun111/Pre_hospital_mortality).”

Reviewer #2 (Remarks on code availability):

I am unable to access the GitHub repository and the web server, which are intended to provide the code and model for application.

Response> This revised response replaces our earlier reply.

Response to Final Revision

We sincerely appreciate the opportunity to submit the final revised version of our manuscript entitled “Prehospital Real-Time AI for Trauma Mortality Prediction: A Multi-Institutional and Multi-National Validation Study” (Manuscript ID: NCOMMS-25-24191B). We are grateful to the editor and reviewers for their insightful comments and constructive feedback, which have greatly contributed to improving the clarity and scientific rigor of our work.

In this final revision, we have carefully addressed all remaining points raised in the previous round of review. All changes made in response to the comments are clearly indicated in the revised manuscript and a detailed point-by-point response is provided below.

We believe that the revised version now meets the journal’s standards and we respectfully submit it for your consideration.

REVIEWERS' COMMENTS

Reviewer #1 (Remarks to the Author):

Thank you for your thorough and considerate responses to the review comments. I have no additional revisions to request at this time.

Response> Thank you for your valuable comment. We truly appreciate the reviewer’s thoughtful insights and careful evaluation of our manuscript

Reviewer #1 (Remarks on code availability):

Nothing especially.

Response> Thank you for your valuable comment.